

Chakra

See where your carbon leaks.

Close the loop with numbers that pay.

HackOut'26 · Circular Carbon Ecosystem · PS10

Industrial Emission Leak-Point Detector & Circular Alternative Recommender

10

INDIAN SECTORS

30

CIRCULAR INTERVENTIONS

31

CITED EMISSION FACTORS

~40%

FOOTPRINT BEHIND A POSITIVE CASE

Complete project report · product definition, research, methodology and working prototype

Generated 12 September 2026

Contents

01 Chakra

02 Chakra — Executive Summary

03 Chakra — Product Requirements Document

04 Research Dossier

05 Users and Personas

06 Competitive Landscape

07 Innovation, Feasibility and Risk

08 Real-World Impact Model

09 Architecture and Modules

10 Scalability and Business Model

11 48-Hour Execution Plan

12 Pitch Script and Judge Q&A

13 Methodology and Limitations

14 Full-Stack Architecture

15 The Data Flywheel

16 API Reference

17 Complete Feature Catalogue

18 Team Task Allocation — 4 People

19 Security, Tenancy and Testing

Chakra

See where your carbon leaks. Close the loop with numbers that pay.

HackOut'26 · Circular Carbon Ecosystem · **PS10 — Industrial Emission Leak-Point Detector & Circular Alternative Recommender**

An Indian industrial SME enters ten numbers it already has — the electricity bill, the coal purchase, the cotton invoices. Chakra returns a Scope 1/2/3 inventory with uncertainty bands, finds where carbon is leaking **relative to that plant's own sector peers**, and produces a ranked, costed portfolio of circular interventions on a marginal abatement cost curve.

It leads with rupees and closes with tonnes, because an SME owner is not buying a sustainability product — they are buying a cost-reduction product with a compliance side-benefit.

Run it

```
pip install -r requirements.txt
cd prototype/backend
python -m uvicorn app:app --reload --port 8080
```

Open **http://127.0.0.1:8080/**. **No internet connection is needed at any point** — no CDN, no external API. The database is created on first run at `prototype/backend/chakra.db`.

```
python -m pytest tests/test_stack.py -q      # 20 tests
```

Two surfaces

	Anonymous sandbox	Signed-in product
Stores	Nothing	Plants, assessments, actions
Benchmarks	Literature priors	Blended with the live corpus
Tracking	—	Implementation tracker with actuals
Account	Not required	Required

Same engine, same charts. An SME who has just been asked for carbon data gets the full assessment on a plant from their own cluster before handing over anything.

Result on the hero demo — Tirupur knitwear dyeing unit

Annual footprint	24,069 tCO₂e (19,414–29,492, ±20.9%)
Scope split	28% S1 · 8% S2 · 64% S3
Largest leak point	Purchased cotton yarn — 60.6% , critical
Cash-positive portfolio	17 interventions · ₹4.66 Cr/yr · ₹4.56 Cr capex · 12-month payback
Abatement at no net cost	25% of footprint (42% available in total)

Across all ten sectors, roughly **40% of an SME's footprint sits behind a positive business case.**

What makes it more than a calculator

1. **Benchmark-relative leak detection** — a leak is not a big number, it is a *bigger number than the plant next door achieves on the same product*. Three rules: benchmark breach vs sector p75, material concentration, structural Scope 3 hotspot.
2. **A real MACC** — bar width is tonnes, height is ₹/tCO₂e, sorted cheapest first. Everything below the zero line pays for itself.
3. **It refuses** — the engine evaluates and then *rejects* inadmissible interventions. It will not tell a GMP pharma plant to use recycled blister foil, nor a Morbi tile kiln to burn briquettes, and it caps recycled cotton at 25% because staple length falls with every recycling pass.
4. **Interaction de-rating** — seven interventions target the electricity meter; each applies to what the previous one left behind, not to the original bill.
5. **Uncertainty carried to the headline** — every factor has a band, and the band survives every multiplication.
6. **A data flywheel that actually runs** — sector benchmarks shrink from literature toward measured percentiles at $n/(n+8)$. A plant is never benchmarked against its own data, and only its latest assessment counts.
7. **Realisation measured, not assumed** — the impact model has to assume a 25% realisation rate. The action tracker records estimated vs achieved and replaces that assumption with a number.

Layout

docs/	the written work – 19 documents
├─ 00-EXECUTIVE-SUMMARY.md	
├─ 01-PRD.md	product requirements
├─ 02-RESEARCH-DOSSIER.md	market, regulation, factors, benchmarks
├─ 03-USERS-AND-PERSONAS.md	
├─ 04-COMPETITIVE-LANDSCAPE.md	including where we would lose
├─ 05-INNOVATION-AND-FEASIBILITY.md	including the risk register
├─ 06-IMPACT-MODEL.md	
├─ 07-ARCHITECTURE-AND-MODULES.md	the engine
├─ 08-SCALE-AND-BUSINESS-MODEL.md	
├─ 09-EXECUTION-PLAN-48H.md	
├─ 10-PITCH-AND-QA.md	demo script + hostile-question prep
├─ 11-METHODOLOGY-AND-LIMITATIONS.md	
├─ 12-FULLSTACK-ARCHITECTURE.md	the platform layer
├─ 13-DATA-FLYWHEEL.md	how benchmarks become measured
├─ 14-API-REFERENCE.md	all 20 routes
├─ 15-FEATURE-CATALOGUE.md	everything built, and what isn't
├─ 16-TEAM-TASKS-4-PEOPLE.md	full task split for a team of four
├─ 17-SECURITY-AND-TESTING.md	tenancy model + what the tests prove
prototype/	
├─ backend/	
│ ├─ app.py	FastAPI, 20 routes, two surfaces
│ ├─ db.py	sqlite3 schema, 6 tables
│ ├─ auth.py	bcrypt + revocable server sessions
│ ├─ repo.py	org-scoped data access
│ ├─ benchmarks.py	live corpus blending + realisation stats
│ ├─ report.py	per-assessment PDF, server-rendered MACC
│ ├─ tests/test_stack.py	20 full-stack tests
│ └─ engine/	
│ ├─ constants.py	NCVs, tariffs, CRF, thresholds
│ ├─ factors.py	units + uncertainty bands
│ ├─ footprint.py	Scope 1/2/3 inventory by stream
│ ├─ leaks.py	three detection rules
│ ├─ macc.py	matching, economics, de-rating, refusal
│ └─ assess.py	orchestration, Sankey, compliance
├─ frontend/	zero dependencies, hand-built SVG
└─ data/	
├─ emission_factors.json	31 factors + 15 state grids, sourced, banded
├─ interventions.json	30 circular interventions
└─ sectors.json	10 Indian sectors + benchmarks + demo plants
build/	

└─ Chakra-Report.pdf	the full project report
└─ Chakra-Plant-Report.pdf	a sample per-assessment report

Reference data is JSON so a domain expert can extend it without touching Python.

Status

Working end to end. **20/20 tests pass**; all 10 sectors assess with **zero invariant violations** (stream sums, abatement ceilings, de-rating monotonicity, substitution caps).

What it is not

A **screening tool**, not a BEE-accredited energy audit and not an assurance engine. Its economics are planning-grade estimates meant to rank options and justify getting a vendor quotation — not to replace one. Every limitation is listed in [docs/11-METHODOLOGY-AND-LIMITATIONS.md](#) and [docs/15-FEATURE-CATALOGUE.md](#), and surfaced in the product's own UI.

Chakra — Executive Summary

HackOut'26 · Circular Carbon Ecosystem · PS10 Industrial Emission Leak-Point Detector & Circular Alternative Recommender

Chakra — see where your carbon leaks, close the loop with numbers that pay.

The one-paragraph version

India has roughly 63 million MSMEs. A growing number of them are being asked — by EU customers under CBAM, by listed Indian customers under BRSR Core, by lenders under green finance criteria — to state their carbon footprint and show a reduction plan. Almost none can. The tools that exist are enterprise carbon accounting platforms priced for corporates with sustainability departments, and they answer the wrong question: they tell you *what* your footprint is, not *what to do about it, what it costs, and when it pays back*. Chakra takes ten minutes of data an SME owner already has on their electricity bill and purchase ledger, finds the points where carbon is leaking relative to their own sector's peers, and returns a ranked, costed portfolio of circular interventions on a marginal abatement cost curve — leading with rupees, closing with tonnes.

The problem, stated precisely

An SME factory owner in Tirupur or Coimbatore faces three simultaneous facts:

1. **They do not know where their carbon comes from.** Intuition says "the boiler". Often it is the purchased cotton or steel, which is 60–70% of the footprint and invisible on any utility bill.
2. **Generic advice does not convert.** "Install solar" is not a decision. "₹2.17 Cr capex, 44-month payback, removes 367 tCO₂e/yr" is a decision.
3. **The deadline is commercial, not moral.** CBAM's definitive regime began in January 2026. An exporting foundry has a reporting obligation *now*, and its European customer is the one asking.

What Chakra does

Stage	Output
1. Inventory	Scope 1/2/3 footprint under GHG Protocol, every figure carrying an uncertainty band and a cited factor
2. Leak detection	Streams ranked by three rules — benchmark breach vs sector p75, material concentration, structural hotspot — each with the peer percentile and the tonnes recoverable by reaching median
3. Recommendation	30-intervention circular library matched to the plant, each costed to a levelised cost of abatement (₹/tCO ₂ e), capex, payback and NPV
4. Decision	A marginal abatement cost curve. Everything below the zero line pays for itself.
5. Compliance	Indicative CBAM export exposure and a BRSR disclosure readiness checklist

The three things that make this win

1. **It produces a MACC, not a pie chart.** The marginal abatement cost curve is what actual climate consultants deliver at ₹2–10 lakh per engagement. Bar width is tonnes abated, bar height is ₹ per tonne, sorted cheapest first. Everything below the line is money the plant is currently leaving on the table. No student team will build this, and every judge with industry exposure will recognise it instantly.

2. It says no. The engine evaluates and then *refuses* interventions that are technically valid but inadmissible — it will not tell a GMP pharma plant to use recycled blister foil, and it will not tell a Morbi tile kiln to burn briquettes. It caps recycled cotton at 25% because staple length falls with every recycling pass. A recommender that never says no cannot be trusted when it says yes, and this is the single clearest signal to a judge that the team understands the domain rather than the API.

3. It is honest about its own uncertainty. Every emission factor carries a low/base/high band, and the band propagates to the headline: *24,069 tCO₂e (range 19,414–29,492, ±20.9%)*. The methodology panel lists limitations rather than hiding them. This is the difference between a demo and a tool an auditor could actually use.

Demonstrated result — Tirupur knitwear dyeing unit

Annual footprint	24,069 tCO₂e (19,414–29,492, ±20.9%)
Scope split	28% Scope 1 · 8% Scope 2 · 64% Scope 3
Largest leak point	Purchased cotton yarn — 60.6% of footprint, <code>critical</code>
Benchmark breaches	Electricity at p78, process heat at p76 of sector peers
Cash-positive portfolio	17 interventions · ₹4.66 Cr/yr net benefit · ₹4.56 Cr capex · 12-month blended payback
Abatement at no net cost	25% of total footprint ; 42% available in total
Correctly refused	Nothing blocked in this sector; recycled cotton capped at 25% by spinning constraint

The same engine runs ten Indian industrial sectors, each with its own benchmarks, demo plant and constraint set.

Status

Working end-to-end: Python engine (6 modules), FastAPI service (7 endpoints), zero-dependency SVG frontend with hand-built Sankey and MACC. Ten sector profiles assessed clean with zero invariant violations. Runs entirely offline — no external API can fail during a demo.

What it is not

It is a **screening tool**, not a BEE-accredited energy audit and not an assurance engine. Its economics are planning-grade estimates meant to rank options and justify getting a vendor quotation — not to replace one. It says so on every card, and it says so to judges.

Read next: `01-PRD.md` for the full product definition, or `10-PITCH-AND-QA.md` for the three-minute demo script and the hostile-question prep.

Chakra — Product Requirements Document

Version 1.0 · Status Built and demonstrable · Problem statement HackOut'26 PS10

1. Problem definition

1.1 The problem statement, as given

Small and medium industries often don't know exactly where their carbon footprint originates or what circular alternatives are available. This problem asks for a tool where a business inputs process data (energy source, materials, waste streams) and the system identifies top emission sources, then recommends specific circular interventions (e.g., alternative materials, recycling loops, process changes) along with estimated cost and CO2 savings.

1.2 The reading that matters

Two readings of "leak-point detector" are possible and only one is correct.

- ✗ **Literal:** a sensor system detecting physical gas leaks. This is not what the PS asks for — it says the business *inputs process data*, not that the system reads instruments.
- ✓ **Correct:** the point in a process where carbon escapes the business unnecessarily. "Unnecessarily" is the load-bearing word, and it is only meaningful **relative to what comparable plants achieve**.

This single interpretation choice drives the entire architecture. A detector built on the literal reading needs hardware nobody has. A detector built on the correct reading needs **sector benchmarks**, which is a data problem we can solve.

1.3 Why existing behaviour fails

Actor	What they do today	Why it fails
SME owner	Nothing, or a one-off consultant audit at ₹2–10 lakh	Cost; 6–12 week turnaround; result is a PDF that goes in a drawer
Large customer	Emails a spreadsheet to their supplier	Supplier cannot fill it; data returned is guessed or blank
Consultant	Excel model rebuilt per client	Not scalable; no benchmark corpus; no reproducibility
Enterprise SaaS	Sells to the customer, not the supplier	₹40L+/yr, assumes a sustainability team exists

The gap is a **self-serve, sector-aware, prescriptive** tool at SME price and SME effort.

2. Goals and non-goals

2.1 Goals

#	Goal	Measure of success
G1	Produce a defensible Scope 1/2/3 inventory from data an SME already has	≤ 10 input fields per stream; every output carries a source and a band
G2	Identify <i>where</i> carbon leaks, relative to peers	Every leak carries a rule, a severity, a percentile and a tonnes-to-median figure
G3	Recommend costed circular interventions	Every recommendation carries ₹/tCO ₂ e, capex, payback, NPV and a physical statement
G4	Make the decision obvious	One MACC; everything below the zero line is cash-positive
G5	Connect carbon to a commercial deadline	CBAM exposure and BRSR readiness computed, not asserted
G6	Be trustworthy	Refuse inadmissible recommendations; propagate uncertainty; publish limitations

2.2 Explicit non-goals

- **Not** a BEE-accredited energy audit. Screening grade, stated as such.
- **Not** an assurance or verification engine. It produces working papers; a licensed assurer signs them.
- **Not** a carbon credit issuer or trading platform.
- **Not** an IoT/sensor platform in v1. Meter integration is a v2 wedge, not a v1 requirement.
- **Not** a generic global tool. India-first by design: CEA grid factors, Indian fuel prices, Indian cluster benchmarks, INR throughout, Indian regulatory hooks.

3. Users

Summarised here; full personas in [03-USERS-AND-PERSONAS.md](#).

Persona	Role	Primary job-to-be-done	Success looks like
Rajesh — SME owner/MD	Decision maker	"My EU buyer wants carbon numbers and I have two weeks"	A credible number and a plan he can show, in one sitting
Priya — plant/works manager	Implementer	"Which of these do I start on Monday?"	A ranked shortlist filtered to difficulty ≤ 2 and payback < 2 yr
Anand — sustainability consultant	Power user / channel	"I need to do 20 of these a month, not 2"	Reproducible engine, exportable working papers
Meera — corporate ESG lead (customer)	Demand creator	"I need Scope 3 data from 300 suppliers"	Consistent, comparable, auditable supplier submissions
Regulator / lender	Verifier	"Is this claim credible?"	Cited factors, stated uncertainty, declared exclusions

4. Functional requirements

FR-1 — Plant profile intake

- **FR-1.1** Sector selection from a fixed taxonomy of 10 Indian industrial sectors.
- **FR-1.2** State selection, driving the Scope 2 grid emission factor.
- **FR-1.3** Activity capture across five stream families: electricity (kWh/yr), fuels (native units per fuel), materials (t/yr), waste (t/yr), freight (tonne-km/yr).
- **FR-1.4** Scale context: annual output (t), revenue (₹ Cr), headcount.
- **FR-1.5** Economic overrides: electricity tariff (₹/kWh), cost of capital (%), EU export share (%).
- **FR-1.6** A prefilled, realistic demo profile per sector, so a first-time user sees the output before doing any typing.

FR-2 — Footprint engine

- **FR-2.1** Compute Scope 1, 2 and 3 per GHG Protocol Corporate Standard.
- **FR-2.2** Every factor carries `low / base / high`; the band propagates through every multiplication and addition to the headline total.
- **FR-2.3** Unit resolution is read from the factor's own unit string; an unrecognised unit raises rather than silently returning a wrong number.
- **FR-2.4** Thermal fuels aggregate into one addressable `thermal_fuel` stream; diesel is held separate because in an Indian SME it is genset electricity, not process heat, and has entirely different abatement options.
- **FR-2.5** Biogenic CO₂ (biomass combustion) is reported as a separate memo line and **excluded** from the Scope 1 total, per GHG Protocol.
- **FR-2.6** Emit both gate-to-gate (Scope 1+2) and cradle-to-gate intensities. Only gate-to-gate is benchmarked.

FR-3 — Leak-point detection

Three rules, evaluated in priority order, each stream reported once:

- **FR-3.1 Benchmark breach** — stream intensity exceeds sector p75. Severity scales with distance above p75, weighted by the stream's share of total footprint.
- **FR-3.2 Material concentration** — stream is >15% of footprint **and** above sector median.
- **FR-3.3 Structural hotspot** — stream is >25% of footprint with no applicable benchmark (typically a purchased material).
- **FR-3.4** Each leak reports: rule, severity (`critical / high / moderate / watch`), peer percentile, and tonnes recoverable by reaching sector median.
- **FR-3.5** An overall peer position on the gate-to-gate intensity metric.

Why Rule 3 exists. The most common real finding for an Indian engineering SME is that its own plant is efficient and its purchased steel is the problem. A detector that only checked energy intensity would issue a clean bill of health to a plant whose footprint is 70% Scope 3.

FR-4 — Circular intervention recommender

- **FR-4.1** A library of 30 interventions across five categories: energy, material, process, waste, logistics.
- **FR-4.2** Each intervention declares its target stream, abatement fraction (low/base/high), capex basis, savings model, lifetime, difficulty, disruption days, confidence and evidence.
- **FR-4.3** Sector applicability filtering.

- **FR-4.4 Blocking** — named (sector, intervention) pairs are evaluated and refused with a stated reason, surfaced in the UI.
- **FR-4.5 Capping** — named pairs have their substitutable share reduced with a stated reason.
- **FR-4.6** Material substitutions respect a declared physical blend ceiling (`max_substitution_pct`) and the tonnage the plant actually buys.

FR-5 — Economics

- **FR-5.1** Four savings models, because "you save the whole bill" is only true sometimes:
 - `avoided_purchase` — the plant genuinely stops buying that energy
 - `tariff_delta` — the plant still buys it, at a different unit price
 - `fuel_switch` — equal-energy replacement; **the delta can be negative**
 - `price_delta` — material substitution on tonnes switched
 - `none` — the entire cash case sits in the opex delta
- **FR-5.2** Capex annualised by capital recovery factor $CRF = r(1+r)^n / ((1+r)^n - 1)$, never charged in full to year one.
- **FR-5.3** $LCOA = (CRF \cdot capex + \Delta opex - gross\ saving) / annual\ abatement$
- **FR-5.4** Simple payback, NPV over asset life, and a cash-positive flag.
- **FR-5.5 Interaction de-rating** — in MACC order, each intervention applies to the *residual* stream, not the original baseline. Both standalone and de-rated abatement are shown.

Why de-rating matters. Seven separate interventions target the electricity meter. Their standalone fractions sum to over 100% of the bill, which is physically impossible. Once a VFD has removed 8%, the LED retrofit's 3% applies to what remains. This is the first thing an energy auditor checks on a stacked savings claim.

FR-6 — Portfolio views

- **FR-6.1** `all` — every applicable intervention.
- **FR-6.2** `cash_positive_only` — $LCOA < 0$. The headline number.
- **FR-6.3** `quick_wins` — $payback \leq 2$ years **and** $difficulty \leq 2$. The "start Monday" list.
- **FR-6.4** Each with count, abatement, capex, net annual benefit, blended payback and NPV.

FR-7 — Visualisation

- **FR-7.1** Sankey: stream → scope → total, ordered by scope then size to minimise ribbon crossing; tail streams bundled.
- **FR-7.2** MACC: width = tonnes, height = ₹/tCO₂e, zero line marked, axis clamped to the informative band with off-scale bars visibly clipped and labelled.
- **FR-7.3** Leak cards with an inline peer-distribution bar showing p50, p75 and the plant's position.
- **FR-7.4** Expandable working for every recommendation — target stream, abatement range, de-rating, capex, savings, LCOA, NPV, savings model.

FR-8 — Compliance

- **FR-8.1** CBAM indicative exposure = (Scope 1 + Scope 2) × EU export share × reference price.
Precursor/upstream emissions **excluded and declared as excluded**.
- **FR-8.2** BRSR readiness checklist with per-item status and the explicit note that assurance is out of scope.
- **FR-8.3** Sector regulatory flags surfaced.

FR-9 — Transparency

- **FR-9.1** Methodology panel listing standard, GWP set, grid factor source, biogenic treatment.
- **FR-9.2** A "limitations we are not hiding" section, open by default, covering factor provenance, verification status, benchmark caveat, leak rule, de-rating limits and capex grade.

5. Non-functional requirements

ID	Requirement	Rationale
NFR-1	Zero external runtime dependencies in the browser	Venue wifi fails. Both charts are hand-built SVG; no CDN.
NFR-2	Stateless API	The plant's data never persists. A better answer to "where is the data stored" than an unused database.
NFR-3	Assessment completes in < 300 ms	It must feel like a calculator, not a batch job.
NFR-4	Every numeric output traceable to a cited factor	Auditability is the product.
NFR-5	Runs offline end-to-end	Demo safety.
NFR-6	Reference data is JSON, not code	A domain expert can extend the library without touching Python.

6. Data model

```
PlantProfile
├─ name, sector, state
├─ annual_output_t, annual_revenue_cr, employees
├─ electricity_kwh          → Scope 2
├─ fuels{FUEL_KEY: qty}    → Scope 1  (native units per fuel)
├─ materials{MAT_KEY: t}   → Scope 3
├─ waste{WASTE_KEY: t}     → Scope 3
├─ freight{MODE_KEY: t·km} → Scope 3
└─ tariff, discount_rate, eu_export_share_pct

Assessment
├─ headline      – the four numbers and the plain-English statement
├─ footprint     – scopes, streams, intensities, biogenic memo, uncertainty
├─ sankey        – nodes + links
├─ leaks         – ranked findings with peer position
├─ recommendations – ranked list, MACC curve, three portfolios, blocked list
├─ compliance    – CBAM, BRSR
└─ methodology   – standard, sources, limitations
```

Reference data: `emission_factors.json` (31 factors across 5 groups, plus 15 state grid variants), `interventions.json` (30), `sectors.json` (10 sectors with benchmarks + demo profiles).

7. Acceptance criteria

#	Criterion	Status
AC-1	All 10 sectors assess without error	✓ 10/10
AC-2	Stream emissions sum to the reported total	✓ verified
AC-3	No intervention abates more than its target stream holds	✓ verified
AC-4	De-rated abatement never exceeds standalone	✓ verified
AC-5	Substitution never exceeds the declared blend cap	✓ cotton capped at 25%
AC-6	Blocked interventions appear in the UI with a reason	✓ pharma rPET, ceramics briquette
AC-7	Capped interventions state the restriction	✓ food rPET 35%, auto steel 45%
AC-8	Fuel switch to a more expensive fuel shows as a net cost	✓ biomass at +₹1,630/tCO ₂ e
AC-9	Headline carries an uncertainty band	✓ ±20.9% on hero demo
AC-10	Page renders with network disabled after first load	✓ no external assets

8. Out of scope for v1, sequenced for v2

Status update. Items 1 (partially), 5 and 7 have since been built, along with accounts, persistence and implementation tracking. See `12-FULLSTACK-ARCHITECTURE.md` and `15-FEATURE-CATALOGUE.md` for what actually exists.

1. Multi-site rollup and year-on-year tracking
2. Utility bill OCR / DISCOM portal ingestion to remove manual entry
3. Smart meter and SCADA integration for measured rather than declared baselines
4. Vendor marketplace — connect a recommendation to three quotes
5. Lender pack export for SIDBI / green term loan applications
6. Supplier-portal mode for a corporate to onboard its supplier base
7. Benchmark corpus built from real assessments, replacing literature percentiles

Research Dossier

Everything the product rests on, with provenance and an honest confidence rating.

Verification discipline. Figures below are drawn from published literature and public policy documents as understood at build time. Anything marked ⚠️ must be re-verified against the primary source before commercial use. A hackathon build is allowed to use literature values; it is not allowed to pretend they are primary measurements.

1. Market

1.1 Size of the addressable problem

India has on the order of **63 million MSMEs**, contributing roughly **30% of GDP** and **~45% of exports**. Industrial MSMEs — the subset with a process, a boiler or a furnace, and a material bill — are the target. Even a conservative slice of registered manufacturing MSMEs is a market in the hundreds of thousands of plants.

The relevant number is not "how many SMEs exist" but **how many are under carbon pressure today**. Three forcing functions, in descending order of immediacy:

Forcing function	Who it hits	Timing	Confidence
CBAM — EU Carbon Border Adjustment Mechanism	Exporters of iron & steel, aluminium, cement, fertiliser, electricity, hydrogen	Transitional reporting ran Oct 2023–Dec 2025; definitive regime from Jan 2026 ⚠️ verify current phase and certificate price	High
BRSR / BRSR Core — SEBI	Top listed companies, cascading to their value chain i.e. SME suppliers	Phased; value-chain disclosure already driving supplier data requests	High
Green finance — SIDBI, bank ESG lending	SMEs seeking term loans for efficiency capex	Ongoing	Medium

The wedge is CBAM. It is the only one of the three with a legal deadline, a monetary penalty, and a customer actively demanding the data. An Indian foundry exporting castings to Germany has a problem this quarter. That is a buyer, not a prospect.

1.2 Industrial clusters targeted

Cluster concentration is what makes SME distribution tractable — one association meeting reaches 200 plants running near-identical processes.

Cluster	Sector	Why it matters
Tirupur, TN	Knitwear dyeing	Export-heavy, already under ZLD regulation, buyer pressure from EU brands
Coimbatore / Rajkot / Belgaum	Foundry	CBAM-exposed ; high electricity intensity; aluminium substitution upside
Morbi, GJ	Ceramic tiles	Extremely thermal-intensive; gas-fired; hard-to-abate showcase
Ludhiana, PB	Light engineering, hosiery	Steel-heavy Scope 3; poor grid reliability → diesel
Ankleshwar / Vapi, GJ	Chemicals	Solvent recovery upside; hazardous waste cost
Pune / Chakan, MH	Auto components	OEM customers cascading BRSR Core requests

2. Regulatory research

2.1 CBAM (EU Regulation 2023/956)

- Covers **embedded emissions** in imported goods across cement, iron & steel, aluminium, fertilisers, electricity and hydrogen. ⚠️ Verify the current Annex I CN code list — scope has been revised and may expand.
- Transitional period (reporting only) **1 Oct 2023 – 31 Dec 2025**; definitive regime with certificate surrender **from 1 Jan 2026**. ⚠️ Confirm current status.
- Embedded emissions comprise **direct** emissions plus, for some goods, **indirect** (electricity) emissions, plus **precursor** emissions.
- Requires **installation-level** data. Default values are available but are deliberately punitive relative to actual verified data — which is precisely the commercial incentive to measure.

Design consequence. Chakra computes exposure as $(\text{Scope 1} + \text{Scope 2}) \times \text{EU export share} \times \text{reference price}$ and **explicitly excludes precursor emissions**, because precursor rules require supplier-specific data the tool does not hold. Guessing it and presenting it as a number would be the exact failure mode the product exists to prevent. The UI says this in the panel.

2.2 BRSR and BRSR Core (SEBI)

- BRSR is a mandated sustainability disclosure for the largest listed Indian companies.
- BRSR Core** specifies a subset of KPIs requiring **reasonable assurance**, and extends disclosure expectations into the **value chain**.
- Mechanism of relevance: an unlisted MSME is never directly regulated — but its listed customer needs value-chain numbers, so the request arrives as a **purchase-order condition**, not a regulation.

Design consequence. The BRSR panel is a readiness checklist, and it is explicit that Chakra produces working papers, **not assurance**. Claiming assurance would be a material misrepresentation.

2.3 Other Indian instruments referenced

Instrument	Relevance
CEA CO ₂ Baseline Database	Source of the Indian grid emission factor; regional variation drives Scope 2
PAT scheme (BEE)	Source of documented Indian industrial energy savings; evidence base for several interventions
Plastic Waste Management Rules	EPR registration and recycled content targets — makes rPET substitution a compliance move, not just a carbon move
Fly ash utilisation notification	Legal basis for the mature ash-to-cement symbiosis market
Hazardous Waste Rules (CPCB)	Used foundry sand and solvent disposal liability
Solid Waste Management Rules	Authorised recycler framework underpinning waste segregation
IS 1489 / IS 455	Blended cement standards — the technical basis for the substitution being permissible
IEC 60034-30-1	Motor efficiency classes (IE1–IE5)

3. Emission factor research

31 factors across five groups, plus 15 state grid variants. Each carries `value` , `low` , `high` , `unit` , `scope` and `source` .

3.1 Electricity — the highest-leverage single choice

Factor	Value	Band	Source
India national grid	0.716 tCO ₂ e/MWh	0.680–0.760	CEA CO ₂ Baseline Database ⚠️ re-check latest version
State variants	0.48 (KL) – 0.88 (WB)	—	Derived from CEA regional mixes

Finding: state choice can move a Scope 2 result by ~40%. West Bengal runs roughly 1.8× Kerala per unit consumed. Any tool using a single national number for a country this electrically heterogeneous is introducing a larger error than most of its other assumptions combined. **This is why state is a required field.**

The national factor is **declining year on year** as renewable share grows — which means a Scope 2 reduction claim must state its vintage or it is not comparable across years.

3.2 Fuels

Fuel	Factor	Unit	Band	Note
Diesel	2.68	kgCO ₂ e/l	2.60–2.75	DEFRA stationary combustion
Natural gas	2.02	kgCO ₂ e/Sm ³	1.92–2.12	DEFRA gross CV
LPG	2.94	kgCO ₂ e/kg	2.88–3.00	DEFRA
Furnace oil	3.15	kgCO ₂ e/kg	3.05–3.25	IPCC 2006 Vol.2 residual fuel oil
Indian coal	1.70	tCO₂e/t	1.45–2.00	IPCC adjusted for Indian ash/NCV
Biomass briquette	0.06	tCO ₂ e/t	0.02–0.12	CH₄/N₂O only

Finding — Indian coal has the widest band in the entire database (±16%). Indian non-coking coal is graded G1–G17 with wildly varying ash content and net calorific value. A tool that uses a single global "coal" factor for an Indian plant is making its largest error on its largest Scope 1 stream. The band is carried through rather than hidden.

Finding — biogenic carbon is a trap. Under GHG Protocol, biogenic CO₂ from biomass combustion is reported **separately** and excluded from the Scope 1 total; only CH₄ and N₂O from combustion are counted. Many student tools set biomass to zero with no explanation, which is *numerically* close but *methodologically* indefensible and immediately visible to anyone who knows the standard. Chakra computes the biogenic memo line (≈1.55 tCO₂/t) and discloses it.

3.3 Materials — where the circular economy actually lives

Material pair	Primary	Secondary	Reduction
Aluminium	13.0 tCO ₂ e/t	0.60	~95%
Steel	2.20	0.55	~75%
Cotton yarn	5.50	1.80	~67%
PET	3.00	1.30	~57%
Cement (OPC→blended)	0.85	0.55	~35%
Paper/board	1.25	0.80	~36%
Glass	1.10	0.70	~36%

Finding — recycled aluminium is the single largest circular lever available to an Indian SME. ~95% embodied carbon reduction *and* a lower price per tonne. It is cash-positive and carbon-massive simultaneously. Any foundry or auto-component plant buying primary aluminium has this sitting unexploited.

Counter-finding — the ceiling is specification, not willingness. Alloy trace-element limits, staple length in recycled cotton, food-contact rules for rPET, and OEM material certification for safety-critical steel all cap the substitutable share far below 100%. This is why `max_substitution_pct` exists, and why a model without it produces recommendations the process physically cannot accept.

3.4 Transport and waste

Rail freight is ~0.022 kgCO₂e/t-km against road at ~0.095 — roughly **4× better per tonne-km**. Modal shift is a large, low-capex Scope 3 lever for bulk inbound, blocked by lead time and safety stock rather than cost.

Organic waste to landfill at 0.58 tCO₂e/t is dominated by **CH₄ from anaerobic decay** under AR6 GWP100. Anaerobic digestion scores **negative** (−0.12) because it both avoids that methane and displaces fossil gas — it counts twice.

4. Benchmark research

Sector intensity percentiles (p25/p50/p75) across ten sectors on three metrics: electricity kWh/t, thermal GJ/t, and gate-to-gate tCO₂e/t.

 **Honest status:** these are **indicative screening percentiles** compiled from published sector energy-intensity literature and cluster study ranges. They are adequate to rank a plant and flag outliers. They are **not** a BEE-accredited benchmark set, and the product says so on the leak panel.

4.1 The basis error we found and fixed

Our first benchmark set mixed **cradle-to-gate** values (some sectors) with **gate-to-gate** values (others) and compared both against a cradle-to-gate computed intensity. The textile demo came out at 8.02 tCO₂e/t against a "p75" of 3.4 — reporting a perfectly ordinary plant as catastrophically bad.

Resolution: benchmark **only** gate-to-gate (Scope 1+2) intensity, because that is what published sector energy benchmarks measure and it is the only part the plant directly controls. Cradle-to-gate is reported alongside but never benchmarked — a plant that buys more material per tonne of product is not thereby running a worse factory. Scope 3 dominance is surfaced through the structural-hotspot rule instead.

This is the most important methodological correction in the build, and it is worth stating to judges because **mixing the two bases is the single most common error in SME carbon tools**.

5. Intervention evidence base

30 interventions, each with a confidence rating and an evidence note.

Confidence	Count	Meaning
high	13	Well-evidenced across many Indian SME deployments
medium	15	Evidenced but strongly site-sensitive
low	2	Promising, thin Indian evidence base

5.1 Selected findings

Compressed air leaks are the cheapest tonne in almost every factory. Industrial systems typically lose 20–30% of compressor output to leaks, and compressed air is commonly 10–15% of plant electricity. Ultrasonic survey plus tagged repair pays back in weeks with zero production stoppage. Modelled at 4% of plant electricity (2.0–6.5%).

Open access is a carbon move, not a cash move. Renewable procurement via open access can cut Scope 2 by 25–80% at near-zero capex — but the plant *still buys the electricity*, so the cash case is only the tariff differential, and cross-subsidy plus additional surcharge can erase it entirely. Modelling it as "half your electricity bill disappears" overstates the business case by roughly 20×. This drove the `savings_model1` design.

Fuel switching to biomass costs money. Coal at ₹6,500/t and 15.9 GJ/t is ~₹409/GJ. Briquette at ₹7,200/t and 15.0 GJ/t is ~₹480/GJ. The switch is **~₹71/GJ more expensive**. It remains the largest Scope 1 lever available to a

thermal-heavy SME — and it is not free. Reporting it as free would be the most consequential lie the tool could tell.

Behavioural savings regress. Idle-load elimination is nearly free and delivers ~5% of electricity, but it decays within 6–9 months without sub-metering and a named owner. Captured as a caveat rather than quietly ignored.

6. Open research questions

#	Question	Impact if wrong	Mitigation in v1
R1	Are the sector benchmark percentiles representative of actual Indian clusters?	Peer percentiles misleading	Labelled as indicative; caveat on panel; replaceable JSON
R2	Current CBAM certificate price and CN code coverage	Exposure figure off	Reference price is a named constant; caveat requires user confirmation
R3	Latest CEA grid factor vintage	Scope 2 off by a few %	Band carried; source named
R4	Are recycled-material price deltas stable?	Cash case shifts	Prices in JSON, user-overridable; flagged as volatile for briquette
R5	Real blend ceilings per alloy/grade	Over- or under-recommending substitution	Conservative caps; stated on the card

None of these are hidden. Every one appears in the product's own limitations panel. A tool whose stated uncertainties match its actual uncertainties is more useful than one that projects false precision — and considerably harder to take apart in Q&A.

Users and Personas

The PS names four user groups: **SMEs, factory operators, sustainability consultants, industry regulators**. Below, each is expanded into who they actually are, what they are trying to do, and what the product must therefore do. Two adjacent users the PS does not name are included because they are where the money and the distribution come from.

Primary users

P1 — Rajesh · SME owner / Managing Director

The buyer. The one whose signature releases capex.

Context	35–55, runs a ₹20–80 Cr turnover plant, often second-generation. Deeply knows the process, does not know GHG Protocol and has no reason to.
Trigger	An email from his European customer asking for embedded emissions data, with a deadline. Or a bank asking for an ESG declaration on a term loan.
Current behaviour	Forwards it to whoever seems least busy. Gets a quote from a consultant for ₹3 lakh and 8 weeks. Stalls.
Job to be done	"Give me a number I can defend and a plan I can afford, before I lose this customer."
Fears	Looking incompetent to his customer. Spending capex on something with no return. Being told to do something that breaks his process.
What he does NOT care about	Tonnes of CO ₂ , in the abstract. Climate framing. Certificates.

Product consequences - Lead with **rupees**, close with tonnes. The headline KPI is ₹4.66 Cr/yr not 6,013 tCO₂e. - Every recommendation must carry **payback in months**, because that is the unit he thinks in. - The output must be **showable** — something he can forward to the customer who asked. - Never recommend anything that would break the process. This is why **blocking** exists; one bad recommendation ends his trust permanently.

P2 — Priya · Plant manager / Works engineer

The implementer. The one who decides whether anything actually happens.

Context	Engineer, 10–20 years on the floor. Knows exactly which compressor leaks and which trap has been failed for two years. Has no budget authority and no spare hours.
Trigger	Rajesh forwards the report and says "look into this".
Job to be done	"Which two of these can I start on Monday without stopping production, and what do I tell the boss it costs?"
Fears	Downtime. Being blamed when a vendor over-promises. Another initiative that dies in three months.

Product consequences - The **quick wins** portfolio exists entirely for Priya: payback ≤ 2 years **and** difficulty ≤ 2. - Every intervention carries **disruption_days** — she needs to know whether it takes the line down. - Every

intervention carries a **physical statement**: "Avoids about 120,000 kWh a year". She validates against her own intuition before she believes the rupees. - **Caveats are a feature, not a disclaimer**. "Savings regress within 6–9 months without sub-metering" is exactly the sentence that earns her trust, because she has watched it happen.

P3 — Anand · Sustainability consultant

The power user and the highest-leverage distribution channel.

Context	Runs a 3–10 person practice, or is a solo BEE-certified energy auditor. Currently rebuilds an Excel model per client.
Trigger	Wins an engagement; needs to produce a credible report faster than the fee justifies at current effort.
Job to be done	<i>"Do twenty of these a month instead of two, without my margin collapsing or my name going on a number I can't defend."</i>
Fears	A tool that makes him look like he used a toy. Numbers he cannot trace to a source when the client's auditor asks.

Product consequences - Every figure must be traceable to a cited factor. This is non-negotiable for Anand and it is why the reference endpoint exists. - Assumptions must be **overridable** — his client's actual tariff, actual cost of capital. - The engine must be **reproducible**: same inputs, same outputs, every time. No hidden randomness, no model drift. - He is the channel: one consultant brings 20 plants. Pricing and packaging should reflect that (see [08-SCALE-AND-BUSINESS-MODEL.md](#)).

P4 — Regulator / auditor / lender

The verifier. Does not use the tool; judges its output.

Context	State pollution control board officer, BRSR assurer, or a SIDBI credit officer assessing a green term loan.
Job to be done	<i>"Is this claim credible enough to accept, or do I have to send it back?"</i>
Fears	Accepting a number that later proves fabricated.

Product consequences - Uncertainty bands on every headline. A number without a range reads as false precision to this reader. - **Declared exclusions**. "CBAM precursor emissions are excluded because we do not hold supplier-specific data" is far more credible than a complete-looking number that quietly guessed. - **Explicit scope statement** — what is counted and what is not, per GHG Protocol. - The product must never claim **assurance**. It produces working papers; a licensed assurer signs them.

Secondary users (not named in the PS, but decisive)

S1 — Meera · Corporate sustainability lead at a large customer

The demand creator. The reason Rajesh opens the email at all.

She has 300 suppliers and a BRSR Core value-chain obligation. She sends a spreadsheet; most come back blank or obviously guessed. She cannot use inconsistent data.

Product consequences — this is the **B2B2B distribution wedge**. If Meera's company hands Chakra to its supplier base, acquisition cost per SME approaches zero and the data comes back in one consistent, comparable

format. It also creates the benchmark corpus. This is the single most important growth mechanism in the model.

S2 — Cluster association / MSME-DFO officer

Tirupur Exporters' Association, an induction furnace association, a District Industries Centre. One meeting reaches 200 near-identical plants. **Cluster concentration is why SME distribution is tractable in India at all** — and it is why sector templates matter more than a generic form.

Non-users, stated deliberately

Who	Why not
Large listed corporates	Already served by enterprise carbon SaaS with sustainability teams. We would lose and should not try.
Individual consumers	Different product entirely.
Carbon credit buyers/traders	We do not issue, verify or trade credits.
Service businesses (offices, IT)	No process, no material bill, no leak to find. The whole method assumes a factory.

The user insight that shapes everything

An SME owner is not buying a sustainability product. He is buying a cost-reduction product with a compliance side-benefit.

Every design decision follows from this:

Decision	Because
MACC sorted by ₹/tCO ₂ e, not by tonnes	He optimises cash, not carbon
Headline KPI is annual rupee benefit	It is the number he repeats to his CFO
Payback shown in months under two years	Months is his unit; years feels like "someday"
"Quick wins" portfolio filtered by difficulty	Priya needs a Monday list, not a strategy
Cash-positive interventions separated out	<i>"Do these regardless of your view on climate"</i> is the only pitch that survives a sceptical owner
CBAM panel	Converts a nice-to-have into a deadline with a rupee figure attached

A tool that opens with "reduce your carbon footprint" loses this user in four seconds. A tool that opens with *"₹4.66 crore a year, 12-month payback, and it happens to remove a quarter of your emissions"* keeps him for the whole meeting.

Competitive Landscape

"Does this already exist?" is the question that kills most hackathon projects. The honest answer here is: **the accounting layer exists and is crowded; the prescriptive layer for SMEs does not.** This document says exactly where we sit and where we would lose.

⚠️ Competitor positioning is based on public market understanding at build time. Pricing is indicative. Verify before using any specific claim commercially.

1. The landscape in four tiers

Tier 1 — Enterprise carbon accounting platforms

Persefoni, Watershed, Sweep, Plan A, Normative, Sphera

What they do	Full-scope carbon accounting, supplier engagement, disclosure-grade reporting (CSRD, SEC, CDP)
Who they sell to	Large corporates with a sustainability function
Price	Enterprise annual contracts, typically far beyond SME reach
Onboarding	Weeks, usually with implementation support
Strength	Audit-grade rigour, deep integrations, genuine assurance workflows
Why they don't serve our user	Priced and designed for an organisation with a dedicated team. An SME owner will never see a demo, let alone a contract.
Do they prescribe?	Partially — mostly target-setting and scenario modelling, not costed plant-level interventions with Indian payback

Tier 2 — ESG ratings and supply-chain assessment

EcoVadis, CDP Supply Chain

Assess and **score** suppliers. They are the source of the pressure our user feels, not the relief from it. A supplier who gets a poor EcoVadis score still has no idea which valve to fix. **Complementary, not competitive** — arguably a demand generator for us.

Tier 3 — Indian carbon / ESG software

A real and growing set of Indian ESG reporting and carbon accounting vendors. ⚠️ This tier is the one to research hardest before making any claim; it is where a genuine overlap is most likely.

Strength	India-aware, BRSR-native, reasonable pricing, local sales motion
Typical focus	Reporting and disclosure workflow — getting the BRSR filed
Gap	Still predominantly <i>accounting</i> : what is my number. Less commonly <i>prescription</i> : which twelve things should I do, in what order, at what ₹/tCO ₂ e

Tier 4 — Consultants and free spreadsheets

Option	Cost	Turnaround	Problem
BEE-accredited energy audit	₹2–10 lakh	6–12 weeks	Cost; result is a static PDF; no material/Scope 3 view
GHG Protocol Excel tools	Free	Days of expert effort	Requires someone who already knows the standard; no benchmarks; no economics
Sector association guidance	Free	—	Generic; not plant-specific; not costed

The consultant is our real competitor and our best channel simultaneously. We do not replace the audit — we make the screening pass 50× cheaper so the audit is spent where it matters.

2. Where we actually differ

Capability	Enterprise SaaS	Indian ESG tools	Consultant	Free XLS	Chakra
Scope 1/2/3 inventory	✓	✓	✓	●	✓
Indian grid factors by state	●	✓	✓	X	✓
Peer benchmarking → leak detection	●	X	●	X	✓
Costed interventions (₹, payback, NPV)	●	X	✓	X	✓
Marginal abatement cost curve	●	X	●	X	✓
Circular material substitution modelling	X	X	●	X	✓
Refuses inadmissible recommendations	X	X	✓ (a good one)	X	✓
Interaction de-rating on stacked savings	●	X	✓	X	✓
Uncertainty bands propagated	●	X	●	X	✓
CBAM exposure estimate	●	●	✓	X	✓
Self-serve in 10 minutes	X	●	X	X	✓
SME-affordable	X	●	X	✓	✓

✓ strong · ● partial/varies · X absent

3. The five defensible claims

We can defend exactly these, and should not overclaim beyond them.

1. We answer "what should I do" rather than "what is my number." The market is saturated with accounting. The output of an accounting tool is a number; the output of Chakra is a **ranked, costed, constraint-checked action list**. That is the difference between a thermometer and a prescription.

2. We produce a MACC for a plant that could never afford one. The marginal abatement cost curve is the standard artefact of professional climate consulting. Delivering it self-serve, in seconds, at SME scale is genuinely new at this price point.

3. We model circular material substitution as a first-class lever, with physical ceilings. Most carbon tools are *energy* tools with a Scope 3 estimate bolted on. For an Indian auto-component or foundry SME, **purchased material is 60–70% of the footprint** and recycled aluminium is a 95% reduction sitting unexploited. We model the substitution, its price delta, *and* the alloy/blend ceiling that caps it.

4. We refuse. No competitor at this price point will tell a pharma plant that recycled blister foil is inadmissible under GMP, or a Morbi kiln that briquettes will not fire glazed tile. A recommender that never says no cannot be trusted when it says yes.

5. India-first, not India-localised. CEA state grid factors, Indian coal grade bands, Indian fuel prices in ₹, cluster-specific benchmarks for Tirupur and Coimbatore and Morbi, CBAM and BRSR hooks, INR and lakh/crore throughout. Not a Western tool with a currency setting.

4. Where we would lose — stated honestly

Against	Why we lose	Response
Enterprise SaaS, for a large corporate	No assurance workflow, no audit trail infrastructure, no integrations, no SOC 2	Don't compete. Different buyer entirely.
A good consultant, on accuracy	They measure; we estimate from declared data	Position as screening before audit , not instead of it
Indian ESG reporting vendors, on disclosure workflow	They do the filing; we do not	Complementary — we produce the reduction plan their report needs
Anyone, on benchmark authority	Our percentiles are literature-derived, not a surveyed corpus	Stated openly in the UI; fixed by the data flywheel (\$5)

The honest summary: we are the best screening-and-prescription layer for an Indian industrial SME, and we are not an audit, an assurance engine, or an enterprise platform. Claiming otherwise would be the fastest way to lose credibility with the one judge in the room who knows this market.

5. The moat, if this became a company

A hackathon prototype has no moat. The path to one:

- Benchmark corpus.** Every assessment improves the sector percentiles. After a few hundred plants in a cluster, *"you use 2.3× more energy per tonne than the median Tirupur dyeing unit"* is a claim nobody else can make. This is a genuine data network effect, and it is the only durable one here.
- Intervention outcome data.** Tracking which recommendations were implemented and what they actually delivered converts literature estimates into measured priors. Nobody else is collecting this at SME scale.
- Channel lock-in via corporates.** Once a large customer standardises its 300-supplier base on one format, switching is organisational, not technical.
- Constraint library.** The blocked/capped rules encode domain knowledge that took real expertise to assemble. It compounds and it is invisible to scrapers.

None of these exist today. Saying so is more persuasive than pretending otherwise.

Innovation, Feasibility and Risk

Part 1 — Innovation

1.1 What is genuinely novel

Not everything here is novel, and claiming otherwise invites a judge to find the weak claim and discard the strong ones with it. Sorted by how defensible the claim is.

#	Claim	Novelty	Defence
I1	Benchmark-relative leak detection	High	Defining a "leak" as intensity above sector p75 — rather than as a large absolute number — makes the detector meaningful without sensors. Three complementary rules catch outliers, large-and-above-median streams, <i>and</i> structural Scope 3 hotspots that intensity alone would miss.
I2	Self-serve MACC at SME scale	High	The marginal abatement cost curve is a professional consulting artefact. Producing one automatically, in seconds, from ten inputs, at SME price is new at this tier.
I3	Constraint-aware refusal	High	An engine that <i>evaluates and then refuses</i> — GMP blocking recycled pharma packaging, kiln chemistry blocking briquettes, staple length capping recycled cotton at 25% — inverts the usual recommender failure mode.
I4	Interaction de-rating	Medium-high	Applying each intervention to the residual stream rather than the original baseline. Standard practice in professional energy auditing; almost never implemented in software at this level.
I5	Savings-model taxonomy	Medium-high	Distinguishing <code>avoided_purchase</code> from <code>tariff_delta</code> from <code>fuel_switch</code> prevents the 20× overstatement that comes from treating "50% of electricity abated" as "50% of the electricity bill saved".
I6	Uncertainty propagated to the headline	Medium	Bands carried through every operation rather than collapsed at the first multiplication.
I7	Circular substitution as a first-class lever	Medium	Modelling primary→secondary material switching with price deltas and physical ceilings, rather than treating Scope 3 as a footnote.
I8	Scope 1/2/3 accounting per GHG Protocol	None	This is table stakes. We do not claim it as innovation.
I9	Sankey and dashboard	None	Presentation, not innovation.

1.2 The central insight

A recommendation is only useful if it is ranked by the thing the user optimises. An SME optimises cash.

Almost every carbon tool ranks by tonnes abated. That ordering is useless to the person who has to sign the cheque, because the biggest-tonnage option is frequently the most expensive one. Ranking by **levelised cost of**

abatement produces a completely different — and actionable — ordering, and it surfaces the finding that matters most:

A quarter of this plant's footprint can be removed at negative cost. Not "should be removed for the planet" — *removed while making money*. That reframing is the product.

1.3 The second insight — saying no is a feature

Recommender systems are optimised to recommend. The failure mode is a plausible-looking suggestion that is inadmissible in context. In an industrial setting one such recommendation destroys trust permanently — an owner who is told to put recycled resin in a GMP blister pack will never open the tool again.

Chakra ships three refusal mechanisms:

- **Blocked** — evaluated, refused, reason shown (*pharma rPET; ceramics briquette*)
- **Capped** — allowed at a reduced share, reason shown (*food rPET 35%; auto steel 45%*)
- **Physical ceiling** — declared per intervention (*recycled cotton 25%, alloy limits, etc.*)

The UI shows these under "**Considered and rejected**". It is the most trust-building screen in the product.

Part 2 — Technical feasibility

2.1 Feasibility rating: very high

PS10 has the best feasibility profile of any problem statement in the set, for one structural reason: **zero external runtime dependencies**. No satellite API, no live grid feed, no IoT stream, no third-party rate limit. Everything needed is public reference data that can be embedded in the repository.

Risk class	Typical hackathon project	Chakra
External API fails on stage	High	None — nothing external is called
Rate limit / auth expiry	High	None
Data licensing	Medium	Public factors, cited
ML training time	High	No model training required
Venue wifi	High	Runs fully offline

2.2 Why no machine learning

The PS suggests "AI/ML recommendation models". We deliberately did not use one, and this is a defensible engineering decision rather than an omission:

1. **No training data exists.** There is no labelled corpus of Indian SME plants and their implemented interventions. A model trained on synthetic data learns the synthesiser's assumptions.
2. **The physics is known.** Emission factors and abatement fractions come from thermodynamics and published LCI, not from pattern-matching. Replacing a correct deterministic calculation with an approximation is a downgrade.
3. **Explainability is a requirement, not a nice-to-have.** An auditor asking "why 412 tonnes?" must get a derivation, not a feature-importance plot.
4. **Where ML genuinely belongs is v2:** learning actual abatement realisation rates from implementation outcomes, and clustering plants for better peer grouping. Both need data we do not yet have.

Stated to judges as: "We use a deterministic, auditable model because carbon accounting has a right answer and a standard that defines it. ML enters when we have implementation-outcome data worth learning from — that is v2, and here is exactly what it would learn."

2.3 Build complexity — actual

Component	Lines	Complexity	Status
<code>factors.py</code> — loading, units, uncertainty	~180	Medium	✓
<code>footprint.py</code> — Scope 1/2/3, streams, intensities	~230	Medium	✓
<code>leaks.py</code> — 3 detection rules, percentiles	~290	Medium	✓
<code>macc.py</code> — matching, economics, de-rating	~430	High	✓
<code>assess.py</code> — orchestration, Sankey, compliance	~250	Low	✓
<code>app.py</code> — FastAPI, 7 endpoints	~140	Low	✓
Frontend — SVG Sankey + MACC, no deps	~480	Medium-high	✓
Reference data — 31 factors, 30 interventions, 10 sectors	~1,100 (JSON)	The real work	✓

The honest assessment: the code is not hard. The *domain data* is where the effort and the differentiation live. Any competent team can write a MACC calculation in four hours; assembling 30 interventions with defensible abatement fractions, capex bases, savings models, physical ceilings and sector constraints is what takes the time and is what a judge cannot replicate by reading the screen.

2.4 Performance

Full assessment for a 10-stream plant against 30 interventions: well under 300 ms, entirely CPU-bound arithmetic. No database, no network, no model inference. Scales linearly; a single small instance handles thousands of assessments per hour.

Part 3 — Risks

3.1 Product and credibility risks

ID	Risk	Likelihood	Impact	Mitigation
R1	A judge with domain expertise challenges a specific emission factor	High	Medium	Every factor carries a band and a named source. Answer: " <i>here is the range and the source; the point estimate is screening-grade and the UI says so.</i> " Never defend a point estimate.
R2	"This is just a calculator"	Medium	High	Demonstrate the three things a calculator cannot do: benchmark-relative detection, de-rating, refusal.
R3	Benchmarks are literature-derived, not surveyed	Certain	Medium	Stated in the UI and in this document. Framed as the v2 data flywheel.
R4	Economics look too good	Medium	High	Already caught and fixed once — the pre-fix model showed ₹10.5 Cr/yr benefit on a ₹34 Cr plant. Savings models and de-rating brought it to a credible ₹4.66 Cr. Telling this story is itself evidence of rigour.
R5	CBAM details out of date	Medium	Medium	Reference price is a named constant; panel carries a "confirm before acting" caveat.
R6	Garbage inputs → confident garbage outputs	High in real use	Medium	v1 accepts declared data. v2 adds bill-based validation and cross-checks against sector plausibility ranges.

3.2 The risk we already hit

During build, the model claimed 18 of 21 interventions were profitable, delivering ₹10.5 Cr/yr of net benefit on a ₹34 Cr revenue plant — 31% of turnover. That is obviously wrong, and three separate bugs caused it:

1. Every energy intervention assumed the whole stream cost disappeared, including open access (where you still buy the power) and fuel switching (where you still buy fuel).
2. Material substitution ignored the physical blend ceilings declared in its own caveat text — recommending a 52% recycled-cotton switch when staple length caps it near 25%.
3. Seven electricity interventions stacked additively to ~99% of the meter.

After fixing all three: **₹4.66 Cr/yr, 12-month blended payback, 25% of footprint** — and the biomass fuel switch correctly flipped from "free" to a **net cost of ₹1,630/tCO₂e**.

This is worth saying out loud in the pitch. "*Our first model said this plant could save ₹10 crore a year. We didn't believe it, we found three modelling errors, and here is what we fixed*" demonstrates more engineering judgement than any feature.

3.3 Ethical risk

Recommending something that damages a business is worse than recommending nothing. Specific guards:

- Difficulty and `disruption_days` on every card — the user sees the operational cost
- Caveats that state failure modes, not just benefits
- Blocking and capping for inadmissible options
- Confidence ratings, including `low` where the Indian evidence base is thin
- No claim of assurance, anywhere

And the inverse: overstating savings to make a demo impressive would, at scale, cause SMEs to commit capex they cannot recover. The corrections in §3.2 were made for that reason, not for the demo.

Real-World Impact Model

The PS states three impact goals. This document takes each one and attaches a computed number rather than an assertion — then states plainly what would have to be true for those numbers to hold.

Impact goals from the PS • Makes emission sources visible and actionable for smaller businesses • Drives adoption of circular practices through concrete, costed recommendations • Supports compliance with emerging carbon regulations

1. Impact at one plant — measured, not assumed

Every figure below is engine output for the Tirupur knitwear dyeing demo profile (2,400 t/yr output, ₹34 Cr revenue, 180 employees).

Metric	Value
Baseline footprint	24,069 tCO ₂ e/yr (range 19,414–29,492)
Scope split	28% / 8% / 64% (S1/S2/S3)
Leak points found	3 — one critical , two moderate
Recoverable by reaching sector median	2,616 tCO ₂ e/yr
Interventions matched	21
Cash-positive subset	17 interventions
Abatement at no net cost	6,013 tCO ₂ e/yr — 25% of footprint
Capex required	₹4.56 Cr
Net annual benefit	₹4.66 Cr/yr
Blended payback	12 months
Total abatement available (incl. net-cost options)	10,210 tCO ₂ e/yr — 42%
Quick wins (payback ≤2 yr, difficulty ≤2)	7 interventions · ₹25.9 L capex · ₹56.7 L/yr · 5-month payback

Interpretation: the plant can remove a quarter of its carbon while *making* ₹4.66 crore a year, and can start on ₹25.9 lakh of quick wins that pay back in five months. The remaining 17% of available abatement costs money — and the tool says so rather than burying it.

1.1 Across all ten sectors

Sector	Footprint tCO ₂ e	Cash-positive abatement	Net benefit ₹/yr	Payback
Textile dyeing	24,069	25.0%	₹4.66 Cr	12 mo
Foundry / casting	18,786	57.9%	₹5.51 Cr	13 mo
Ceramics	26,075	28.4%	₹9.08 Cr	13 mo
Food processing	7,623	60.6%	₹3.92 Cr	16 mo
Auto components	11,807	48.3%	₹2.66 Cr	13 mo
Pharma formulation	7,127	50.3%	₹3.55 Cr	22 mo
Plastic moulding	8,053	37.6%	₹1.90 Cr	17 mo
Paper / packaging	13,960	36.9%	₹4.34 Cr	10 mo
Fabrication	5,169	51.1%	₹1.41 Cr	12 mo
Chemicals	7,458	36.0%	₹1.89 Cr	23 mo

Median cash-positive abatement across sectors: 43% of footprint, at a blended payback of 12–13 months.

That consistency is itself a finding: across ten different Indian industrial processes, roughly **40% of the carbon is sitting behind a positive business case that nobody has computed.**

2. Scaled impact — with the arithmetic shown

 **This section is a model, not a measurement.** Every assumption is named so it can be argued with.

2.1 Assumptions

Assumption	Value	Basis / confidence
Mean footprint per target SME	13,000 tCO ₂ e/yr	Mean of the ten demo profiles. Medium — demo plants are mid-sized; the real long tail is smaller.
Cash-positive abatement identified	40% of footprint	Median across sectors. Medium-high — engine output.
Realisation rate	25%	Fraction of identified cash-positive abatement actually implemented within 2 years. Low confidence — this is the load-bearing assumption.
Effective abatement per plant	$13,000 \times 0.40 \times 0.25 =$ 1,300 tCO₂e/yr	

2.2 Scenarios

Horizon	Plants assessed	Abatement realised	Equivalent to
Hackathon → pilot	50	65,000 tCO ₂ e/yr	—
Year 1, two clusters	500	650,000 tCO ₂ e/yr	~140,000 cars off the road
Year 3, ten clusters	5,000	6.5 M tCO ₂ e/yr	—

And the economic mirror: at ~₹3.5 Cr median net annual benefit per plant, 500 plants implementing a quarter of their identified portfolio is on the order of **₹440 Cr/yr of retained margin in the Indian MSME sector.**

2.3 The assumption most likely to be wrong

The 25% realisation rate. Identifying an opportunity is not implementing one. Real barriers:

- Capital access — even a 12-month payback needs the 12 months of capital up front
- Attention — the owner has a business to run
- Vendor trust — a good recommendation with a bad contractor fails
- Behavioural regression — idle-load savings decay without metering

If realisation is 10% rather than 25%, every number above falls by 60%. It would still be a worthwhile product. Stating this is more credible than defending 25%.

What would raise realisation — and is therefore the v2 roadmap: lender pack export (solves capital), vendor marketplace with three quotes (solves trust), and implementation tracking (solves attention and regression).

3. Impact against each stated PS goal

Goal 1 — "Makes emission sources visible and actionable for smaller businesses"

Before	After
"Our carbon is probably the boiler"	24,069 tCO ₂ e, decomposed into 8 streams with shares and uncertainty
No comparison	Peer percentile on three metrics; 3 leak points with severity
No sense of size	64% is Scope 3 — the opposite of the owner's intuition

The single most valuable output is often a surprise: for the auto-component and foundry demos, the plant's own operations are efficient and its **purchased steel or aluminium is the problem.** No energy-only tool would ever tell them that.

Goal 2 — "Drives adoption of circular practices through concrete, costed recommendations"

Circular interventions in the library: recycled aluminium and steel, rPET, recycled board, blended cement, recycled cotton, in-house regrind, foundry sand reclamation, solvent recovery, closed-loop dye water, anaerobic digestion, industrial symbiosis, waste segregation.

Each carries: tonnes abated, capex, net annual benefit, ₹/tCO₂e, payback, NPV, difficulty, disruption days, confidence, evidence, caveats, and a physical statement of what changes.

The conversion mechanism is the negative LCOA. "Switch to recycled aluminium" is an appeal. "Switch 448 tonnes of aluminium, save ₹1.79 Cr/yr and 5,560 tCO₂e, capex ₹11 lakh" is a purchase order.

Goal 3 — "Supports compliance with emerging carbon regulations"

- **CBAM** — indicative exposure in ₹, with precursor emissions explicitly excluded and declared
- **BRSR** — readiness checklist with per-item status and an explicit statement that assurance is out of scope
- **Sector flags** — ZLD, EPR, hazardous waste, fly ash utilisation surfaced per sector

Compliance is the wedge, not the product. It gets the meeting; the MACC closes it.

4. Co-benefits not counted in the carbon number

Real but deliberately excluded from headline figures, because inflating a carbon claim with adjacent benefits is how impact reporting loses credibility:

Co-benefit	Mechanism
Air quality	Coal→biomass and reduced diesel cut PM and SOx locally — the health benefit accrues to the workers and the neighbourhood
Water	Closed-loop dye water cuts freshwater draw in water-stressed Tirupur
Landfill diversion	Waste segregation and AD divert tonnage from municipal landfill
Industrial symbiosis	Fly ash and slag become another industry's input — two footprints fall
Export competitiveness	A CBAM-ready supplier keeps the contract a non-compliant one loses
Worker safety	Steam line insulation removes a genuine burn hazard
Energy security	Genset displacement reduces exposure to diesel price and outages

5. What honest success looks like at 48 hours

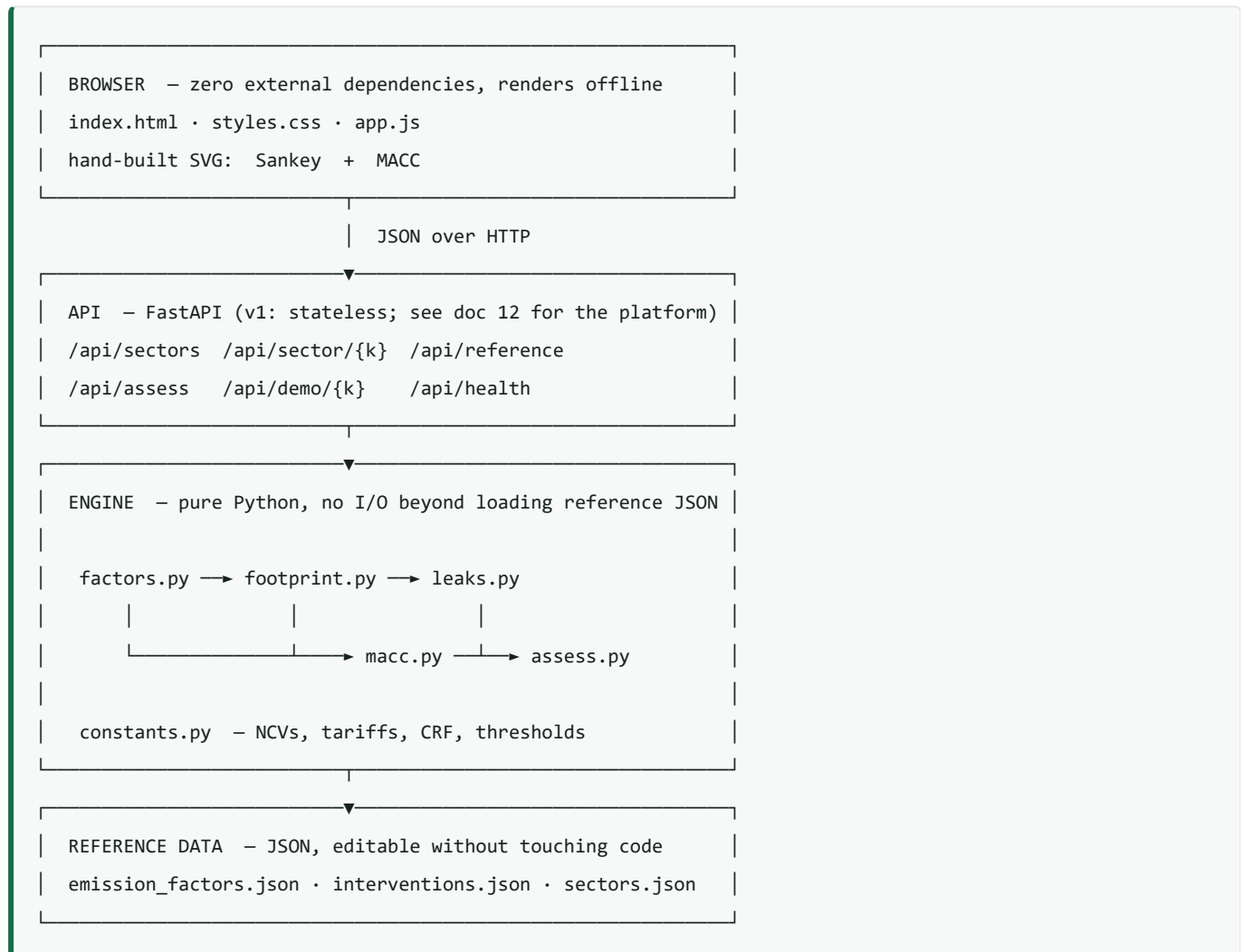
Not "we will decarbonise Indian industry." The credible claim is narrower and stronger:

We built a working tool that takes ten inputs an SME already has, finds where their carbon leaks relative to their own sector, and returns a ranked, costed, constraint-checked plan — demonstrating on ten real Indian industrial sectors that roughly 40% of an SME's footprint sits behind a positive business case nobody has computed for them.

That is a claim we can fully defend with the artefact in the room.

Architecture and Modules

1. System shape



Why the v1 engine is stateless. The engine layer described in this document holds no state: a profile arrives in the request, is assessed, and is returned. That remains true today — `factors`, `footprint`, `leaks`, `macc` and `assess` are pure functions with no database awareness, which is why the anonymous sandbox and the signed-in product cannot drift apart.

Superseded in part. A persistence layer was subsequently added — accounts, stored assessments, implementation tracking and the live benchmark corpus. The engine did not change; it gained an optional benchmark override. See [12-FULLSTACK-ARCHITECTURE.md](#). The anonymous sandbox still stores nothing, so the privacy answer above still holds for it.

2. Modules

`constants.py` — the assumption surface

Every number the model leans on, in one file so it can be audited without reading the maths.

- **NCV table** — net calorific values, to express mixed fuels on a common GJ basis for benchmarking

- **THERMAL_FUELS** — deliberately excludes diesel, which in an Indian SME is genset electricity, not process heat
- **Financial defaults** — ₹8/kWh industrial tariff, 12% cost of capital (MSME term lending is ~11–14%)
- **capital_recovery_factor(r, n)** — $r(1+r)^n / ((1+r)^n - 1)$
- **size_multiplier(revenue)** — banded, not linear: a leak survey at a ₹100 Cr plant is not ten times the job it is at ₹10 Cr

factors.py — units and uncertainty

- **Band** — a (base, low, high) triple with arithmetic that keeps the band. Negative scaling correctly flips which end is the low end.
- **FactorDB** — flattens all five factor groups into one lookup so callers need not know which group a key lives in
- **Unit resolution** — the numerator is read off the factor's unit string (kgCO2e/... → ×0.001, tCO2e/... → ×1). An unrecognised unit **raises** rather than silently returning a wrong number.
- **grid_factor(state)** — state lookup with the national band width carried onto the state point value

footprint.py — the inventory

Produces **Stream** objects, not just scope totals, because a scope total is not actionable.

- Scope 2 from electricity × state grid factor
- Scope 1 from combustion, with **thermal fuels aggregated** into one addressable stream (abatement acts on process heat as a whole) and **diesel held separate**
- **Biogenic CO₂** computed as a memo line and **excluded** from the Scope 1 total per GHG Protocol
- Scope 3 from materials, waste and freight
- **Two intensity bases** emitted: gate-to-gate (Scope 1+2) and cradle-to-gate. Only gate-to-gate is benchmarked.

leaks.py — detection

Three rules, priority-ordered, each stream reported once (a stream benchmarked on two bases must not raise two findings).

Rule	Trigger	Severity driver
benchmark_breach	intensity > sector p75	distance above p75, weighted by stream share
material_concentration	>15% of footprint and > p50	distance above p50 × share
structural_hotspot	>25% of footprint, no benchmark	share

_percentile() places the plant on the peer distribution by piecewise-linear interpolation across p25/p50/p75, clamped to 1–99 so the UI never claims perfection or total failure.

macc.py — matching and economics

The densest module and the one worth reading.

Target resolution maps an intervention's declared target to a live stream, including the aggregates

material_ALL and **waste_all**.

Five savings models — the fix for the single largest class of error:

Model	Meaning
avoided_purchase	The plant genuinely stops buying that energy
tariff_delta	Still buys it, at a different price — only the differential counts
fuel_switch	Equal-energy replacement; the delta can be negative
price_delta	Material substitution on tonnes actually switched
none	Whole cash case sits in <code>opex_delta</code>

Substitution ceilings — capped by `max_substitution_pct`, by any `SECTOR_CAPS` override, and by the tonnage the plant actually buys.

BLOCKED / `SECTOR_CAPS` — the refusal layer. Blocked pairs are evaluated then refused with a stated reason; capped pairs proceed at a reduced share with the restriction shown.

Economics

```
annualised_capex = CRF(r, lifetime) × capex
LCOA = (annualised_capex + Δopex - gross_saving) / abatement
payback = capex / (gross_saving - Δopex)
NPV = -capex + Σ net_benefit / (1+r)^y
```

Interaction de-rating — after sorting by LCOA, each intervention applies to the *residual* of its target stream:

```
remaining[stream] = 1.0
for rec in sorted_by_lcoa:
    frac = rec.abatement / rec.stream_total
    rec.portfolio_abatement = rec.stream_total × remaining[stream] × frac
    remaining[stream] *= (1 - frac)
```

Both standalone and de-rated figures are reported. **Known limitation, declared in the API response:** cross-stream overlap (e.g. in-house regrind against a material substitution) is not de-rated.

`assess.py` — **orchestration**

Composes footprint → leaks → recommendations, then adds: - **Sankey** — three-level `stream → scope → total` graph - **Compliance** — CBAM exposure `(S1+S2) × export share × reference price`, with precursors excluded *and declared excluded*; BRSR readiness checklist - **Headline** — the four KPIs and a generated plain-English statement - **Methodology** — standard, GWP set, factor provenance, benchmark caveat, leak rule

3. API

Method	Route	Purpose
GET	/api/health	Counts of sectors, interventions, factors
GET	/api/sectors	Sector list with clusters
GET	/api/sector/{key}	Full sector spec incl. benchmarks and demo profile
GET	/api/reference	Every emission factor with source, band, price — powers the methodology view
POST	/api/assess	The product. Profile in, full assessment out
GET	/api/demo/{key}	One-call assessment of a sector's demo profile — what the stage demo hits, so nothing depends on typing
GET	/	Frontend

4. Frontend

Constraint: no CDN, no framework, no build step. Venue wifi fails, and a hackathon demo that depends on unpkg is a demo that dies.

Both charts are hand-built SVG:

Sankey — streams ordered by **scope first, then size**, which eliminates most ribbon crossing (ordering purely by size forces every large Scope 3 stream to cross the whole diagram). Tail streams bundle into "N smaller sources". Ribbons are cubic Béziers with per-scope colour.

MACC — bar width = tonnes, height = ₹/tCO₂e, zero line emphasised. **Axis clamped** to a robust band derived from the 8th–92nd percentile of LCOA values: a single narrow bar at –₹28,000/t would otherwise compress every other intervention into an unreadable sliver. Off-scale bars are drawn to the clamp and **marked with a dashed stroke plus a count annotation**, so a clipped bar never reads as if that were its true value.

Leak cards carry an inline peer-distribution strip showing p50, p75 and the plant's position.

Every recommendation row expands to show the full working — target stream, abatement range, de-rating, capex, gross saving, opex delta, LCOA, NPV, savings model, caveats and restrictions.

5. Extension points

To add	Edit	Code change
An emission factor	emission_factors.json	None
An intervention	interventions.json	None
A sector + benchmarks + demo plant	sectors.json	None
A refusal or cap rule	BLOCKED / SECTOR_CAPS in macc.py	3 lines
A new savings model	macc.py _evaluate	One branch

Reference data is JSON by design so a domain expert — the person whose knowledge actually differentiates this product — can extend it without touching Python.

6. Running it

```
cd prototype/backend  
python -m uvicorn app:app --reload --port 8080
```

Then open `http://127.0.0.1:8080/`. Requires `fastapi`, `uvicorn` and `bcrypt` (see `requirements.txt`); the engine itself is standard library only.

Scalability and Business Model

1. Technical scalability

The engine is pure CPU arithmetic with no I/O and no model inference (persistence sits outside it, in the platform layer). A full assessment — 10 streams against 30 interventions with uncertainty propagation and de-rating — completes in well under 300 ms.

Dimension	Current	Ceiling	Blocker
Assessments/hour, single small instance	—	~10,000+	None meaningful
Sectors	10	Unbounded	Benchmark data per sector
Interventions	30	Unbounded	Domain research per intervention
Emission factors	31 (+15 state grids)	Unbounded	Sourcing and verification
Concurrent users	Stateless → horizontal	Unbounded	None

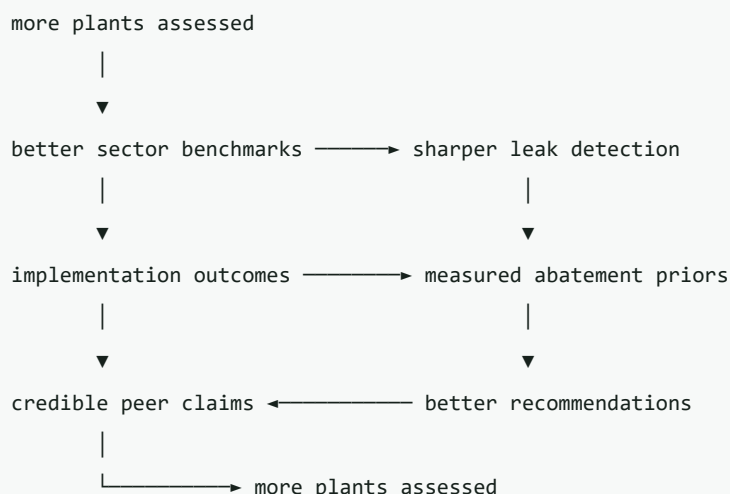
The real scaling constraint is not compute. It is domain data. Adding a sector means assembling defensible intensity percentiles and a demo profile; adding an intervention means researching an abatement fraction, capex basis, savings model and physical ceiling. That is analyst work, not engineering work — which is exactly why it is defensible.

1.1 What breaks first at real scale

#	Breaks	When	Fix
1	Literature benchmarks lose credibility against real cluster data	~100 plants in one cluster	Replace percentiles with the measured corpus — this is the flywheel, not a problem
2	Manual data entry becomes the bottleneck	Immediately, in real use	Bill OCR / DISCOM portal ingestion
3	Stateless design blocks year-on-year tracking	First repeat customer	Add persistence — but only once there is a reason
4	Intervention library becomes too generic for specialist sectors	~20 sectors	Sector-specific sub-libraries
5	Screening estimates diverge from implemented reality	First implementations	Outcome tracking → measured priors (this is where ML finally earns its place)

2. The data flywheel

The only durable moat available here.



After a few hundred plants in one cluster, "you use 2.3× more energy per tonne than the median Tirupur dyeing unit" becomes a claim no competitor can make and no scraper can copy. Peer comparison is also the strongest behavioural driver in the product — being below median is a status problem, not an environmental one, and status moves faster than conscience.

3. Go to market

Ranked by capital efficiency.

Channel 1 — Corporate supplier cascade (*highest leverage*)

A large customer with a BRSR Core value-chain obligation hands Chakra to its 300 suppliers.

- **Acquisition cost per SME** → ~zero. One enterprise conversation onboards hundreds of plants.
- Data comes back in **one consistent, comparable format** — which is the corporate's actual problem.
- Instantly seeds the benchmark corpus within a real supply chain.
- **Who pays:** the corporate, for the supplier programme.

Channel 2 — Cluster associations

Tirupur Exporters' Association, induction furnace associations, District Industries Centres, MSME-DFOs.

- One meeting reaches 200 plants running near-identical processes.
- Sector templates make the tool immediately relevant rather than generically applicable.
- **Cluster concentration is the structural reason SME distribution is tractable in India.**

Channel 3 — Consultants and BEE-certified energy auditors

Anand (persona P3) brings 20 plants a year. Chakra becomes his screening layer; he sells the depth.

- We do not compete with the audit — we make the screening pass 50× cheaper so the audit is spent where it matters.
- **Who pays:** the consultant, per seat or per assessment.

Channel 4 — Lenders and green finance

SIDBI, bank ESG lending desks. A lender assessing an efficiency term loan needs exactly what Chakra produces: costed abatement with payback.

- **Who pays:** the lender, for origination support — and this channel simultaneously solves the capital barrier that caps the realisation rate in `06-IMPACT-MODEL.md` .

Channel 5 — Direct self-serve

The CBAM-panicked exporter who searches at 11pm. Lowest volume, highest intent, best product feedback.

4. Business model

⚠ Indicative. Pricing is a hypothesis to test, not a finding.

Tier	Who	Price	Includes
Free assessment	Any SME	₹0	Full engine, on-screen results, no export
Report	SME	~₹5–15k per assessment	Branded PDF, BRSR-format extract, CBAM working sheet
Consultant seat	Anand	~₹40–80k/yr	Unlimited plants, white-label export, assumption overrides
Supplier programme	Corporate	Annual contract	N supplier assessments, aggregated Scope 3 dashboard, comparable format
Lender pack	Bank/SIDBI	Per origination	Costed abatement pack formatted for credit assessment

Why free at the base tier. The assessment *is* the acquisition. An SME who sees their own footprint decomposed for the first time is qualified in a way no ad can achieve, and every free assessment feeds the flywheel. The paywall sits at **export**, because export is where the value becomes transferable — the thing Rajesh forwards to the customer who asked.

4.1 Unit economics sketch

Marginal cost per assessment	Effectively zero — sub-second CPU
Cost of a sector	Analyst research to assemble benchmarks + demo profile
Cost of an intervention	Analyst research for abatement, capex, savings model, ceilings
Dominant cost	Domain analysis and factor verification, not engineering

Gross margin is software-like; the investment is a research library that compounds.

5. Roadmap

Phase	Focus	Key additions
Now (48h)	Prove the engine	10 sectors, 30 interventions, 31 factors, MACC, refusal layer, CBAM/BRSR panels
~~v1.1~~	Remove entry friction	✓ save/resume, ✓ PDF export · ✖ bill OCR, DISCOM ingest
v1.2	Make it transferable	BRSR-format export, CBAM working sheet, lender pack
~~v2.0~~	Close the loop	✓ implementation tracking · ✖ vendor marketplace
~~v2.1~~	Earn the benchmarks	✓ shipped — shrinkage blending at $n/(n+8)$, see doc 13
v3.0	Measure, don't declare	Smart meter / SCADA integration; ML on implementation outcomes to learn real realisation rates by intervention, sector and plant size

Note on ML. It appears in v3, not v1, and only where it has something real to learn: the gap between estimated and achieved abatement. Using it earlier would mean training on synthetic data to approximate physics we already compute exactly. That sequencing is a deliberate engineering judgement, not an omission — see [05-INNOVATION-AND-FEASIBILITY.md §2.2](#).

6. What would have to be true

The claims in this document depend on four things we have not proved:

1. **SMEs will enter their own data.** Ten inputs is few, but it is not zero. If they will not, Channel 1 (corporate mandate) becomes the only viable route.
2. **Identified savings convert to implemented savings at a meaningful rate.** Everything in the impact model rests on this.
3. **Someone pays for export.** The free tier is only a strategy if the paid tier converts.
4. **Benchmarks can be earned before credibility is spent.** Literature percentiles work for a pilot; they will not survive a sophisticated customer at scale.

Naming these is not hedging. A business model that cannot state its own load-bearing assumptions has not been thought through.

48-Hour Execution Plan

For a four-person team. The build described in this repository is already past hour 30 — this plan is how to get here, and what to do with the rest of the clock.

Principles

1. **The reference data is the product.** Code is a few hundred lines; the 30-intervention library with defensible economics is the thing a judge cannot replicate from the screen. Staff it accordingly.
2. **The engine must be right before the UI is pretty.** A beautiful chart of wrong numbers loses to an ugly chart of right ones the moment a domain judge asks a question.
3. **Nothing external at runtime.** No CDN, no API, no key. The demo must survive the venue wifi dying.
4. **Freeze at hour 42.** The last six hours are rehearsal, not features.

Roles

Role	Owns
A — Engine	<code>factors</code> , <code>footprint</code> , <code>leaks</code> , <code>macc</code> , <code>assess</code> . The economics are their responsibility.
B — Domain data	<code>emission_factors.json</code> , <code>interventions.json</code> , <code>sectors.json</code> . The highest-value seat on the team.
C — Frontend	SVG Sankey and MACC, cards, layout
D — Integration, docs, pitch	API, demo profiles, the deck, the Q&A prep, and the person who says "no" to scope creep

Hour by hour

H0–H4 · Lock the spine

- **All:** agree the interpretation of "leak point" — *benchmark-relative*, not sensor-based. **Get this wrong and 48 hours go the wrong way.** (30 minutes, whole team, non-negotiable.)
- **A:** `Band` arithmetic and `FactorDB` with unit resolution from the unit string.
- **B:** first 15 emission factors with sources and bands. Electricity + fuels first.
- **C:** page skeleton, KPI tiles, no charts yet.
- **D:** FastAPI skeleton, `/api/health` , repo structure.

✔ **Gate H4:** one fuel quantity converts to tCO₂e with a band, through the API.

H4–H12 · Inventory and first sector

- **A:** `footprint.py` — streams, scopes, biogenic memo, both intensity bases.
- **B:** finish the factor set (31); build the **first** sector with benchmarks and a realistic demo plant.
- **C:** stream table + KPI tiles bound to live API data.
- **D:** `/api/assess` , `/api/sector/{k}` , `/api/demo/{k}` .

✅ **Gate H12:** one real plant profile returns a full Scope 1/2/3 inventory on screen.

H12–H20 · Leak detection

- **A:** three detection rules, percentile placement, severity scoring.
- **B:** remaining nine sectors with benchmarks and demo plants.
- **C:** leak cards with the peer-distribution strip.
- **D: invariant test harness** — stream sum = total; no abatement exceeds its stream; all sectors assess clean. Run it after every change from here.

✅ **Gate H20:** leaks fire correctly on at least three sectors, with different rules triggering.

⚠️ **The trap that will cost you four hours here:** mixing cradle-to-gate and gate-to-gate intensity bases between your benchmarks and your computed values. Benchmark **only** Scope 1+2. We hit this and it reported an ordinary plant as catastrophic. See `02-RESEARCH-DOSSIER.md` §4.1.

H20–H30 · The MACC — protect this window

This is the product. Nothing else gets touched until it works.

- **A:** target resolution, savings models, CRF, LCOA, payback, NPV, **interaction de-rating**.
- **B:** all 30 interventions with abatement fractions, capex bases, savings models, ceilings, caveats, evidence.
- **C:** MACC SVG with clamped axis and clipped-bar marking.
- **D:** portfolio views (`all` / `cash_positive` / `quick_wins`).

✅ **Gate H30:** a MACC renders and **the economics are plausible**.

⚠️ **Do not skip the plausibility check.** Our first run claimed ₹10.5 Cr/yr of benefit on a ₹34 Cr plant. Three bugs: whole-stream savings on tariff-delta and fuel-switch interventions, substitution ignoring its own blend ceiling, and seven electricity interventions stacking to 99% of the meter. **If your cash-positive portfolio exceeds ~15% of revenue, you have one of these bugs.**

H30–H38 · Trust layer

The screens that separate this from a calculator.

- **A:** `BLOCKED` and `SECTOR_CAPS` — and make sure the blocked pairs are actually *candidates*, or the refusal path silently never fires. (Ours didn't, at first.)
- **B:** caveats, confidence ratings, evidence notes on every intervention.
- **C:** "Considered and rejected" panel; expandable working on every row; methodology and limitations section.
- **D:** CBAM and BRSR panels.

✅ **Gate H38:** the tool visibly refuses at least two interventions with stated reasons.

H38–H42 · Polish and freeze

- Sankey ordering (scope, then size) to stop ribbon crossing.
- Number formatting in lakh/crore — an Indian judge reading `₹46,570,000` is doing mental arithmetic instead of listening.
- Run every sector once more; confirm zero invariant violations.

- **Feature freeze at H42. No exceptions.**

H42–H48 · Rehearse

- Full run-through **4x**, timed.
- **Once with wifi off.** If it fails, that is what you just learned.
- Pre-load the hero sector so the first click shows results.
- Assign the Q&A: A takes methodology, B takes domain, D takes market.
- Sleep if any is available. A rested presenter beats one more feature.

Cut list, in order

When you fall behind — and you will — cut in this order and never out of order:

Cut #	What goes	Why it's safe
1	Sectors 6–10	Three well-built sectors demo as well as ten
2	Sankey	The MACC is the hero; the Sankey is support
3	CBAM/BRSR panels	Nice wedge, not the core
4	NPV	Payback and LCOA carry the argument
5	Interventions below 20	Depth beats breadth

Never cut: the MACC, the refusal layer, uncertainty bands, or the invariant tests. Those four *are* the differentiation.

Standing rules

- **Commit after every green invariant run.** The test harness is the most valuable thing D builds.
- **Seed everything.** No live typing during the demo; `/api/demo/{sector}` exists for exactly this.
- **One person owns the demo laptop** and does not code on it after H42.
- **Write down every assumption as you make it.** At hour 46 you will not remember why the discount rate is 12%, and a judge will ask.

Pitch Script and Judge Q&A

The three-minute demo

Rule: the first sentence contains a rupee figure, not a carbon figure. You have about eight seconds before a judge decides whether this is another dashboard.

0:00–0:25 · The problem, through one person

"Rajesh runs a knitwear dyeing unit in Tirupur. Two weeks ago his German buyer asked him for his embedded carbon emissions. He has no idea. A consultant quoted him three lakh rupees and eight weeks. His buyer wants it this month.

There are sixty-three million MSMEs in India and a growing number of them just got the same email."

No slides yet. Just the person.

0:25–0:50 · What we built

"Chakra takes ten numbers Rajesh already has — his electricity bill, his coal purchase, his cotton invoices — and in under a second it tells him where his carbon is leaking, and what to do about it, costed in rupees."

Click the sector chip. Results appear.

"Twenty-four thousand tonnes a year. And here is the first surprise: **sixty-four percent of it is not in his factory at all.** It's the cotton he buys. His boiler — the thing he assumed was the problem — is twenty-seven percent."

Point at the Sankey.

0:50–1:25 · The leak detection

"We don't just add it up. We compare him to his own sector. His electricity intensity is 1,250 kWh per tonne against a Tirupur median of 850 — he's in the worst quarter of his peer group. Same on process heat.

That's what a leak point actually is: not a big number, but a **bigger number than the plant next door achieves on the same product.** Closing those two gaps to median alone is worth 2,600 tonnes a year."

Point at the peer strip on the leak card.

1:25–2:15 · The MACC — the hero moment

"Now the part that makes this a decision instead of a report."

Scroll to the MACC. Pause. Let them look at it.

"This is a marginal abatement cost curve. Every bar is one intervention. **Width is tonnes of CO₂. Height is rupees per tonne.** Sorted cheapest first.

Everything **below this line has a negative cost** — it pays for itself and removes carbon as a side effect.

For Rajesh that's seventeen interventions: **four crore sixty-six lakh a year of net benefit, four and a half crore of capital, twelve-month blended payback — and it removes a quarter of his footprint.** He should do all of it today whether or not he cares about carbon.

And this bar" — *point to the amber* — "is the biomass fuel switch. It's his single biggest carbon cut. **It costs him sixteen hundred rupees a tonne.** We show that too, because a tool that only shows you the free wins isn't telling you how to decarbonise."

2:15–2:45 · The trust moment

Scroll to "Considered and rejected". Switch the sector chip to Pharmaceutical.

"One more thing. This is a pharma plant. Our engine evaluated recycled PET for its packaging — and **refused it**, because primary packaging in product contact needs GMP material qualification.

Switch to ceramics: it refuses the biomass switch, because briquette ash ruins a glazed tile kiln.

A recommender that never says no can't be trusted when it says yes. That's the difference between this and a calculator."

2:45–3:00 · Close

"Ten sectors. Thirty circular interventions. Thirty-one cited emission factors plus a grid factor for every state, every one with an uncertainty band that we carry all the way to the headline.

It runs with the wifi off, because it calls nothing external.

Across all ten sectors, we found the same thing: **roughly forty percent of an Indian SME's carbon is sitting behind a positive business case that nobody has ever computed for them.**"

Judge Q&A

Methodology

"Where did your emission factors come from? Are they accurate?"

Published sources — CEA for the Indian grid, IPCC and DEFRA for combustion, worldsteel and IAI for materials. Every one carries a low/base/high band and the band propagates to the headline, which is why we report 24,069 tonnes with a range of 19,414 to 29,492, $\pm 21\%$. We never show a point estimate without its range. And we flag in the tool that these are literature values that need re-verification against the current source editions before commercial use.

Never defend a point estimate. Always retreat to the band and the source — that is a stronger position, and it is the true one.

"Isn't this just a carbon calculator?"

A calculator tells you your number. Three things here a calculator can't do. One, it compares you to your sector and tells you you're in the worst quartile on electricity — that needs a benchmark corpus. Two, it de-rates interacting interventions, so seven electricity measures don't stack to 99% of your meter. Three, it refuses recommendations that are technically valid but inadmissible. The output isn't a number, it's a ranked capital plan.

"Why no machine learning? The PS suggests AI/ML."

Because there's no training data and the physics is known. There is no labelled corpus of Indian SME plants and their implemented interventions — a model trained on synthetic data just learns our own assumptions back. Emission factors come from thermodynamics and published LCI, not pattern matching. And an auditor asking "why 412 tonnes?" needs a derivation, not a feature-importance plot. ML earns its place in v3, learning the gap between *estimated* and *achieved* abatement from real implementation outcomes — and that's exactly what we'd train it on.

"How do you know the benchmarks are right?"

We don't, fully, and the tool says so on the leak panel. They're indicative screening percentiles from published sector energy-intensity literature — good enough to rank a plant and flag outliers, not a substitute for a BEE audit. Replacing them with a measured corpus from real assessments is the core of our data moat, and it's the first thing that improves with usage.

Rigour

"Your savings numbers look too good."

They did to us too. Our first run claimed ten and a half crore a year on a thirty-four crore revenue plant — thirty-one percent of turnover. We didn't believe it and found three bugs. First, every energy intervention assumed the whole stream cost vanished, including open access where you still buy the power and fuel switching where you still buy fuel. Second, substitution ignored the blend ceilings declared in its own caveats — it was recommending 52% recycled cotton when staple length caps it near 25%. Third, seven electricity interventions stacked additively. After fixing all three: 4.66 crore, and the biomass switch correctly flipped from free to a net cost.

This is the single strongest answer in the set. It demonstrates judgement, not features.

"What stops a user entering garbage and getting a confident wrong answer?"

Nothing in v1, and that's an honest limitation. We accept declared data. v2 adds bill-based validation and cross-checks against sector plausibility ranges — if you claim 200 kWh per tonne in a dyeing unit where p25 is 620, we should challenge it rather than compute on it.

Market

"Doesn't this already exist?"

The *accounting* layer absolutely exists and is crowded — Persefoni, Watershed, and a real set of Indian ESG vendors. They answer "what is my number." Almost none answer "which twelve things should I do, in what order, at what rupees per tonne" for a plant that can't afford a consultant. And none of them model circular material substitution with physical blend ceilings, which for an Indian auto-component SME is 60-70% of the footprint.

"Would an SME owner actually use this?"

That's our biggest open risk and we've named it. Ten inputs is few but it isn't zero. Which is why our highest-leverage channel isn't direct — it's a large customer with a BRSR Core value-chain obligation handing this to three hundred suppliers. Acquisition cost goes to near zero, the data comes back in one comparable format, and that's the corporate's actual problem.

"Who pays?"

The assessment is free — it's the acquisition. The paywall is at *export*, because that's where the value becomes transferable: the PDF Rajesh forwards to the buyer who asked. Plus consultant seats and corporate supplier programmes.

Hostile

"You're telling factories to burn biomass. Isn't that just moving the problem?"

Good question, and it's why we handle biogenic carbon explicitly. Under GHG Protocol, biogenic CO₂ is reported separately and excluded from the Scope 1 total — we compute that memo line rather than silently zeroing biomass, which is what most tools do. We also flag the real caveats: seasonal supply volatility, higher ash handling, and boiler warranty implications. And we show it as a **net cost**, not a free win.

"What if your recommendation damages someone's business?"

That's the risk we designed hardest against. Every card carries difficulty, disruption days in production downtime, a confidence rating, and caveats that state failure modes rather than benefits. We block inadmissible interventions outright. And we never claim assurance — this is a screening tool that tells you where to spend an audit, not a replacement for one.

"Three of your ten sectors show no operational leaks at all. Is your detector broken?"

No — that's the detector working. Those plants are genuinely efficient gate-to-gate, and reporting a clean bill of health would be the wrong answer, so the structural-hotspot rule fires instead and tells them their purchased steel is 60% of their footprint. An energy-only detector would have told them they were fine.

Demo safety checklist

- ☐ Server started and hero sector pre-loaded before you walk up
- ☐ **Run once with wifi disabled** — it should work identically
- ☐ Browser zoom set so the MACC is readable from the back of the room

- [] Second sector (pharma) ready for the refusal moment
- [] Know your own numbers cold: 24,069 · 64% Scope 3 · ₹4.66 Cr · 12 months · 25%
- [] Q&A assigned: engine person on methodology, data person on domain, lead on market
- [] If something breaks: `/api/demo/textile_dyeing` returns the full JSON. Read the numbers off it and keep talking.

Methodology and Limitations

The document an auditor would ask for. Everything the engine does, and everything it does not.

1. Standard and boundary

Accounting standard	GHG Protocol Corporate Accounting and Reporting Standard
GWP set	IPCC AR6, 100-year
Organisational boundary	Single facility, operational control
Reporting period	One year, as declared by the user
Currency	INR throughout; lakh/crore in presentation

1.1 Scope coverage

Scope	Included	Excluded — and declared as excluded
1	Stationary combustion: coal, furnace oil, LPG, natural gas, biomass, diesel	Mobile combustion (owned fleet), fugitive refrigerants, process emissions (e.g. calcination)
2	Purchased electricity, location-based, state grid factor	Market-based method; purchased steam/heat
3	Cat. 1 purchased goods (modelled materials), Cat. 4/9 transport, Cat. 5 waste	Business travel, commuting, capital goods, use phase, end-of-life, leased assets, franchises, investments

Exclusions are declared, never reported as zero. A Scope 3 figure that silently omits eight categories while looking complete is worse than one that names what it covers.

2. Calculation chain

2.1 Activity → emissions

emissions_tCO2e = quantity × factor × numerator_multiplier

The numerator multiplier is **read from the factor's own unit string** — `kgCO2e/...` → 0.001, `tCO2e/...` → 1.0. An unrecognised unit raises an exception rather than silently returning a wrong number. Electricity is the one special case: factors are per MWh, the UI collects kWh, and the conversion happens in the engine so the user never does arithmetic.

2.2 Uncertainty

Every factor carries `low` / `base` / `high`. The band is carried through **every** operation:

- Addition: bands add
- Positive scaling: band scales
- **Negative scaling: the band's ends swap** — a common bug, handled explicitly

Reported headline uncertainty is `(high - low) / (2 × base)`, expressed as ±%. The hero demo reports **±20.9%**.

Why this matters. A tool that prints "412 tCO₂e" is making a claim it cannot support. One that prints "412 tCO₂e (360–470)" is making a claim an auditor can check. Collapsing the band at the first multiplication — which is what most tools do — destroys the only honest thing you can say about a screening estimate.

2.3 Biogenic carbon

Per GHG Protocol, biogenic CO₂ from biomass combustion is **reported separately and excluded from the Scope 1 total**. The factor for agri-residue briquette (0.06 tCO₂e/t) covers **only CH₄ and N₂O**. The engine additionally computes the biogenic CO₂ memo line (≈ 1.55 tCO₂/t) so the disclosure is complete.

Setting biomass to zero with no explanation is numerically close and methodologically indefensible.

2.4 Intensity bases

Two are computed; **only one is benchmarked**.

Basis	Computed	Benchmarked	Why
Gate-to-gate (Scope 1+2) per tonne	✓	✓	What published sector energy benchmarks measure; the part the plant controls
Cradle-to-gate (all scopes) per tonne	✓	✗	A plant buying more material per tonne of product is not running a worse factory

This was a correction made during the build. Our initial benchmark set mixed the two bases and reported an ordinary plant as catastrophic (8.02 against a "p75" of 3.4). Mixing cradle-to-gate and gate-to-gate is the most common methodological error in SME carbon tools.

3. Leak detection

A "leak point" is defined as a stream where carbon escapes **unnecessarily** — and unnecessarily is only meaningful relative to peers.

Rule	Trigger	Severity
benchmark_breach	intensity > sector p75	$\min(1, \text{over}/0.5)$ weighted by stream share, +0.15 floor
material_concentration	share > 15% and intensity > p50	distance above p50, scaled by share
structural_hotspot	share > 25%, no applicable benchmark	scaled by share above 25%

Severity bands: `critical` ≥ 0.60 · `high` ≥ 0.35 · `moderate` ≥ 0.15 · `watch` below.

Each stream reports **once**, on whichever basis the sector benchmarks first — a stream benchmarked both per-tonne and per-crore must not raise two findings.

`gap_to_median` = `stream_tCO2e` × (`actual` - `p50`) / `actual` — the tonnes recoverable by reaching sector median.

4. Intervention economics

4.1 Levelised cost of abatement

$$\text{CRF} = r(1+r)^n / ((1+r)^n - 1)$$
$$\text{LCOA} = (\text{CRF} \times \text{capex} + \Delta\text{opex} - \text{gross_saving}) / \text{annual_abatement}$$

Capex is **annualised**, never charged in full to year one. Charging raw capex makes a 25-year solar asset look worse than a 5-year lighting retrofit, which is simply wrong.

Negative LCOA $\Rightarrow$ the intervention pays for itself.

4.2 Savings models

Model	Formula	Used for
avoided_purchase	stream cost $\times$ fraction	Efficiency, captive solar — you stop buying the energy
tariff_delta	quantity $\times$ price differential	Open access — you still buy it, at a different price
fuel_switch	GJ $\times$ (current ₹/GJ – replacement ₹/GJ)	Biomass switch — can be negative
price_delta	tonnes switched $\times$ (old – new price)	Material substitution
none	0	Cash case carried entirely in Δopex

Modelling open access as `avoided_purchase` overstates its business case by roughly **20×**. This taxonomy exists because we made that error and caught it.

4.3 Substitution ceilings

Tonnes switched is capped by, in order: the declared `max_substitution_pct`, any `SECTOR_CAPS` override, and the tonnage the plant actually buys.

Material	Cap	Reason
Recycled cotton	25%	Staple length falls with each mechanical recycling pass
rPET	50% (35% in food)	Food-contact grade approval
Recycled aluminium	70%	Alloy trace-element limits
Secondary steel	80% (45% in auto)	Fatigue-rated component certification
Blended cement	70%	Structural specification
Recycled board	90%	Near-frictionless for transit packaging

4.4 Interaction de-rating

After sorting by LCOA, each intervention applies to the **residual** of its target stream:

```
remaining[stream] = 1.0
for rec in sorted_by_lcoa:
    frac = rec.abatement / rec.stream_total
    rec.portfolio_abatement = rec.stream_total * remaining[stream] * frac
    remaining[stream] *= (1 - frac)
```

Both standalone and de-rated figures are reported on every card.

Known limitation, declared in the API response: cross-stream overlap is not de-rated. In-house regrind (targeting all materials) and a specific material substitution can double-count at the margin. Within-stream interaction — which is where the large errors live — is handled.

5. Compliance calculations

5.1 CBAM

```
exposure_tCO2e = (Scope 1 + Scope 2) * EU_export_share
indicative_cost = exposure * reference_price
```

Precursor and upstream material emissions are excluded. CBAM precursor rules require supplier-specific data this tool does not hold; estimating it would be a guess presented as a number.

The reference price is a named constant, not a live feed. The panel requires the user to confirm current CN codes and certificate price before acting.

5.2 BRSR

A readiness checklist, not a filing. States explicitly that **assurance is out of scope** — the tool produces working papers; a licensed assurer signs them.

6. Complete limitations register

#	Limitation	Severity	Mitigation / route out
L1	Emission factors are literature values, not primary measurements	High	Bands + sources shown; flagged for re-verification before commercial use
L2	Sector benchmarks start as indicative percentiles, not a surveyed corpus	High → shrinking	Partly addressed. A live corpus now blends measured percentiles in at $n/(n+8)$, with the plant excluded from judging itself and provenance shown in the UI. Literature still dominates any sector below ~8 plants. See 13-DATA-FLYWHEEL.md
L3	Capex figures are screening-grade planning estimates	Medium	Stated on every card; intended to trigger a quotation, not replace one
L4	Abatement fractions are literature ranges, strongly site-sensitive	Medium	Low/base/high carried; confidence rating per intervention
L5	Inputs are declared, not measured or validated	High	v2: bill ingestion and plausibility cross-checks
L6	Cross-stream interaction not de-rated	Low	Declared in API response
L7	Eight Scope 3 categories out of scope	Medium	Declared, never reported as zero
L8	Scope 2 is location-based only	Low	Market-based method needs contractual instruments we don't collect
L9	No process emissions (e.g. cement calcination)	Medium	Material for some sectors; declared
L10	CBAM reference price is a constant	Medium	User confirmation required
L11	Single-year snapshot; no trend or seasonality	Low	v1.1
L12	Behavioural savings assumed persistent	Medium	Caveated on the specific interventions where regression is known
L13	No validation that a recommendation was implementable at this specific site	Medium	Difficulty, disruption days and caveats shown; a site visit remains necessary

7. What this tool is not

- **Not a BEE-accredited energy audit.** It is a screening pass that tells you where to spend one.
- **Not an assurance engine.** No claim of verification.
- **Not a carbon credit issuer, verifier or registry.**
- **Not a substitute for a vendor quotation.** Capex is planning-grade.
- **Not a compliance filing.** The BRSR panel is readiness, not submission.

8. Reproducibility

The engine is fully deterministic. Same inputs → same outputs, always. No randomness, no model inference, no external calls, no time dependence. Every figure traces to a named factor in versioned JSON.

Invariants verified across all ten sectors:

- Stream emissions sum to the reported total
- No intervention abates more than its target stream contains
- De-rated abatement never exceeds standalone abatement
- Substitution never exceeds the declared blend ceiling

Result: 0 violations across 10 sectors.

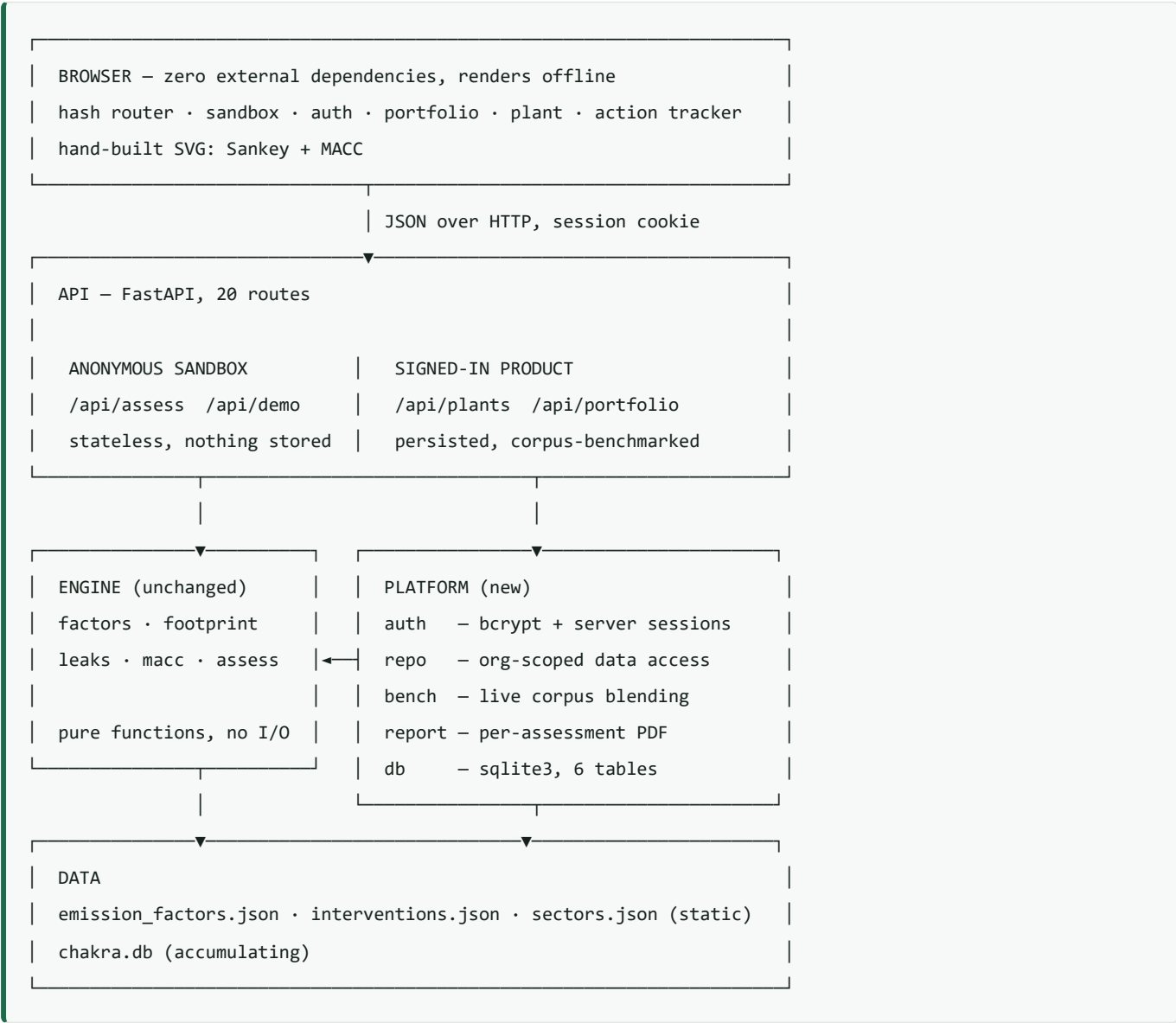
Full-Stack Architecture

The v1 engine was deliberately stateless — profile in, assessment out, nothing written down. That was the right call for a demo and the wrong one for a product, because three of the things that make Chakra defensible only exist once data accumulates:

- 1. Benchmarks that are **measured** rather than assumed
- 2. A realisation rate that is **observed** rather than guessed at 25%
- 3. A plant that can see **whether it improved** year on year

This document covers the layer added to make those real, and what it deliberately did not change.

1. Shape



1.1 The engine did not change

`factors`, `footprint`, `leaks`, `macc` and `assess` remain pure functions with no database awareness. The only addition is an **optional benchmark override**:

```
assess(profile, benchmarks=None, benchmark_provenance=None)
```

Pass nothing and you get literature priors — which is exactly what the anonymous sandbox does. Pass a corpus-blended set and the same engine judges the plant against its real peers. The engine never learns where the percentiles came from; provenance is carried through to the response so the UI can say which was used.

This is why the sandbox and the signed-in product cannot drift apart. There is one engine and one set of chart renderers. A bug in the MACC shows up in both, or in neither.

2. Why two surfaces

	Anonymous sandbox	Signed-in product
Route	<code>/api/assess</code> , <code>/api/demo/{sector}</code>	<code>/api/plants/{id}/assess</code>
Stored	Nothing	Profile + full result, versioned
Benchmarks	Literature priors	Blended with live corpus
Tracking	None	Action tracker with actuals
Needs an account	No	Yes

An SME owner who has just been asked for carbon data by a customer is not going to create an account before he knows whether the tool is useful. He gets the full engine, on a representative plant from his own cluster, with nothing stored and nothing asked of him.

The account is what he creates once he wants to keep the answer.

3. Data model

Six tables. The schema lives in `prototype/backend/db.py` and is created on import; `SCHEMA` is idempotent.

```
orgs          id, name, kind(plant|consultant|corporate), created_at
users         id, org_id→orgs, email UNIQUE, name, pw_hash, role, created_at
sessions      token PK, user_id→users, created_at, expires_at
plants        id, org_id→orgs, name, sector, state, referred_by_org→orgs,
              created_at, archived_at
assessments   id, plant_id→plants, created_at, label,
              profile_json, result_json,
              total_tco2e, scope1/2/3_tco2e,          ← denormalised
              scope12_per_t, elec_per_t, thermal_per_t, ← for corpus queries
              cp_abatement, cp_benefit_inr, cp_capex_inr
actions       id, plant_id, assessment_id, intervention_id, name, category,
              status(recommended|planned|in_progress|done|rejected),
              est_abatement/capex/annual_benefit/payback,
              act_abatement/capex/annual_benefit,      ← what actually happened
              target_date, completed_at, rejected_reason, notes,
              UNIQUE(plant_id, intervention_id)
```

Three decisions worth defending:

Denormalised intensity columns on assessments. The corpus query computes percentiles across a sector's latest assessments. Parsing a JSON result blob per row to reach one float would make the benchmark query cost grow with result size rather than row count. The columns are written once at assessment time.

result_json stored whole. An assessment is a point-in-time statement made under a specific set of emission factors, benchmarks and assumptions. If a factor is corrected next month, last month's report must still say what it said — otherwise it is not an audit trail. So the full result is frozen, not recomputed on read.

UNIQUE(plant_id, intervention_id) on actions. One row per intervention per plant, carried across reassessments. Estimates refresh; status and actuals do not.

3.1 Why sqlite3 and not an ORM

The schema is six tables of flat rows. An ORM would insert a translation layer between what is written and what is stored, and the entire premise of this product is that every number is traceable. Parameterised SQL you can read in one sitting serves that better, and it keeps the dependency surface at zero for the storage layer.

This is a reversible decision. If the schema grows relationships worth mapping, SQLAlchemy is already available.

4. Request lifecycle: a signed-in assessment

POST /api/plants/{id}/assess

1. **auth.current_user** resolves the session cookie against `sessions`, or 401s.
2. **repo.get_plant(org_id, plant_id)** — scoped by org. A plant belonging to another organisation returns 404, not 403: the existence of the row is itself not disclosed.
3. **Identity is overwritten from the plant record**. The submitted profile carries activity data only; `name`, `sector` and `state` come from the database. A client cannot re-sector or rename a plant through the assessment endpoint.
4. **benchmarks.blended_benchmarks(sector, exclude_plant_id=plant_id)** builds the percentile set, excluding this plant's own history.
5. **engine.assess(profile, benchmarks, provenance)** — the unchanged engine.
6. **Persist** the profile, the full result, and the denormalised columns.
7. **_sync_actions** creates or refreshes one action row per recommendation, preserving any status and actuals the user has set, and retiring interventions that no longer apply *unless* the user already acted on them.

Step 7 is the one that matters most for trust. A reassessment must never silently reset someone's implementation tracker — if a plant manager marked eight things done in March, re-running the assessment in June must not quietly wipe that.

5. What was added, file by file

File	Lines	Responsibility
db.py	~170	Schema, connection-per-thread, tiny query helpers
auth.py	~130	bcrypt hashing, server-side sessions, FastAPI dependencies
repo.py	~250	Org-scoped plants, assessments, actions, portfolio rollup
benchmarks.py	~200	Corpus percentiles, shrinkage blending, realisation stats
report.py	~290	Per-assessment HTML + PDF, server-rendered MACC as SVG
app.py	~330	20 routes across both surfaces
tests/test_stack.py	~300	20 tests
Frontend	~560 added	Router, auth, portfolio, plant view, action tracker

6. What is still missing

Stated plainly, because a full-stack claim invites the question.

Gap	Consequence	When it matters
No full activity-data intake form	Reassessment currently edits output, revenue, electricity and export share; fuels/materials/waste/freight come from the sector template	First real user
No email verification or password reset	An account is only as recoverable as the password	First real user
No rate limiting on auth	Brute force is unthrottled	Public deployment
SQLite, single file	One writer at a time	A few hundred concurrent orgs
No CSRF token	Mitigated by <code>SameSite=Lax</code> cookies and a JSON-only API, not eliminated	Public deployment
No migrations tool	Schema changes need care	Second schema change
Sessions never pruned	<code>sessions</code> grows	Housekeeping

None of these block the demo. All of them block a real deployment, and saying so is more useful than a feature list that implies otherwise.

7. Running it

```
pip install -r requirements.txt
cd prototype/backend
python -m uvicorn app:app --reload --port 8080
```

Open `http://127.0.0.1:8080/`. The database is created on first run at `prototype/backend/chakra.db` (gitignored). Set `CHAKRA_DB` to point elsewhere.

```
python -m pytest tests/test_stack.py -q      # 20 tests, throwaway database
```

The Data Flywheel

The single largest limitation of the v1 engine was **L2**: sector benchmarks were literature-derived percentiles, not a measured corpus. This document describes the mechanism that retires it, and — more importantly — why the mechanism is careful rather than clever.

1. The problem

A leak point is defined relative to peers: *your electricity intensity exceeds the sector 75th percentile*. That definition is only as good as the percentile.

Our percentiles came from published sector energy-intensity literature. They are adequate to rank a plant and flag outliers, and they are **not** a surveyed corpus of Indian plants. A sophisticated customer will ask where the number came from, and "a literature range" is a weaker answer than "two hundred plants in your own cluster."

The obvious fix — replace literature with observed data — has an equally obvious failure mode: with four plants assessed, the "75th percentile" is noise wearing a statistic's clothing, and one unusual plant swings the whole sector.

2. The mechanism

Shrinkage toward the literature prior.

$$w = n / (n + N_0)$$
$$\text{blended} = w \cdot \text{empirical} + (1 - w) \cdot \text{literature}$$

$N_0 = 8$, expressed in pseudo-observations. The prior behaves as though it were worth eight real plants.

Plants assessed	Weight on observed data
3	27%
6	43%
12	60%
24	75%
50	86%

The benchmark migrates from assumed to measured **smoothly**. There is no cliff, no version where the sector suddenly re-baselines, and no single plant that can move it far.

Below $\text{MIN_CORPUS} = 3$ the literature value is returned unchanged. Percentiles computed from two plants are not percentiles.

3. The three rules that matter more than the maths

Shrinkage is standard. These are the parts that are easy to get wrong and that quietly destroy the product if you do.

3.1 A plant is never benchmarked against itself

```
blended_benchmarks(sector, exclude_plant_id=plant_id)
```

If a plant's own assessment is in the corpus used to judge it, it drags its own percentile toward its own value. Do that across a sector and every plant looks average, the detector goes quiet, and the tool stops finding anything. This is enforced in the SQL, not in a caller, so there is no path that forgets it.

3.2 Only the latest assessment per plant counts

A plant reassessed twelve times must not outvote eleven plants assessed once. The corpus query takes `MAX(created_at) per plant_id`.

Without this, the most engaged customer silently becomes the benchmark — which is precisely backwards, since the most engaged customer is usually the one improving fastest.

3.3 Provenance travels with the numbers

Every benchmark carries how it was derived:

```
"electricity_kwh_per_t": {
  "source": "blended", "n": 6, "weight": 0.429,
  "literature": {"p25": 780, "p50": 1050, "p75": 1450},
  "observed": {"p25": 812, "p50": 1104, "p75": 1487}
}
```

The UI says *"benchmarked against the live corpus"* or *"benchmarked against literature priors"*, and the leak panel names the sample size. A tool that silently upgrades its own authority as data arrives is doing something the user cannot audit.

4. Observed behaviour

Seeding a nine-plant consultancy portfolio, in order:

#	Plant	Sector	Benchmark source
1	Kovai Castings	foundry	literature
2	Sakthi Alloys	foundry	literature
3	Rajkot Metalworks	foundry	literature
4	Belgaum Foundry Co	foundry	blended
5	Batala Iron Works	foundry	blended
6	Howrah Castings	foundry	blended
7	Tirupur Knits Dyeing	textile	literature
8	Surat Processors	textile	literature
9	Morbi Tile Works	ceramics	literature

The flip happens at plant 4, not plant 3 — because plant 3 sees only two *other* plants once its own data is excluded. That off-by-one is the self-exclusion rule doing its job, and it is the cheapest possible proof that the rule is actually wired in.

Final corpus state:

Sector	Plants	Weight	Status
Foundry / casting	6	0.429	blending in real data
Textile dyeing	2	0.000	literature prior
Ceramics	1	0.000	literature prior

5. The second flywheel: realisation

Benchmarks are one loop. The more valuable one is **estimated versus achieved**.

Every recommendation becomes a tracked action. When someone marks it done, they can record what actually happened:

```
identified    = Σ est_abatement over all actions
committed    = Σ est_abatement where status ∈ {planned, in_progress, done}
realised      = Σ (act_abatement ?? est_abatement) where status = done

realisation_rate    = realised / identified
abatement_accuracy  = Σ act / Σ est    (only where an actual was entered)
capex_accuracy      = Σ act_capex / Σ est_capex
```

5.1 Why this is the most valuable data in the product

`06-IMPACT-MODEL.md` has to **assume** a 25% realisation rate, and openly flags it as the load-bearing assumption — if it is really 10%, every impact number falls by 60%.

This measures it.

On the seeded portfolio, after one completed intervention:

Metric	Value	Meaning
Realisation rate	3.6%	of identified abatement actually delivered
Abatement accuracy	119%	the intervention over-delivered against our estimate
Capex accuracy	83%	it cost less than we projected
Measured n	1	one data point — stated, not hidden

One data point proves the pipeline, not the rate. But the shape of the eventual answer is now a query rather than an argument.

5.2 This is where machine learning finally earns its place

`05-INNOVATION-AND-FEASIBILITY.md` §2.2 argues against ML in v1: no training data exists, and the physics is known.

The realisation table is the training data that did not exist. Once there are enough completed actions, `act / est` becomes a learnable target — realisation by intervention, sector, plant size and difficulty — and the engine can report "solar typically delivers 91% of modelled abatement in this cluster" instead of a literature range.

That is a model trained on outcomes nobody else is collecting, predicting something the physics cannot tell you: not what an intervention *could* do, but what it *does* do when a real SME installs it.

6. Why this is a moat and not a feature

Four properties, in descending order of durability:

1. **It compounds.** Every assessment improves the benchmark for every subsequent plant in that sector.
2. **It is not scrapeable.** The corpus is derived from private plant data that exists nowhere else.
3. **It is self-reinforcing through distribution.** A corporate onboarding 300 suppliers seeds an entire cluster's benchmark in one contract.
4. **It converts a stated weakness into a stated strength over time.** The limitations panel today reads "indicative percentiles". With usage, the same panel reads "blended, 240 plants" — and that sentence is one a competitor cannot write.

7. Honest limits

Limitation	Status
Corpus inherits any bias in who signs up	Early adopters skew toward export-exposed and better-run plants, so early percentiles likely run optimistic
No outlier rejection yet	One mis-keyed assessment shifts a small corpus; shrinkage damps it but does not remove it
No geographic stratification	A Morbi plant and a Khurja plant share one ceramics benchmark, though their gas prices and kiln vintages differ
Sector granularity is coarse	Ten sectors is too few; a jobbing foundry and a high-volume automotive foundry are not peers
<code>N₀ = 8</code> is a judgement call	Defensible, not derived. Worth revisiting against real variance once the corpus is large enough to measure it

All five are visible in the product's own corpus panel or its limitations register. None is hidden behind a confident number.

API Reference

Base URL `http://127.0.0.1:8080` · 20 routes · FastAPI · interactive docs at `/docs`

Authentication is a `chakra_session` cookie, `HttpOnly` + `SameSite=Lax`, set by register/login. Routes marked  require it and return **401** otherwise.

Reference and public

GET `/api/health`

Liveness plus corpus size.

```
{"status":"ok","version":"0.2.0","sectors":10,"interventions":30,
  "factors":31,"plants":9,"assessments":9}
```

GET `/api/sectors`

```
{"sectors":[{"key":"textile_dyeing","label":"Textile dyeing, bleaching and processing",
  "clusters":["Tirupur, TN","Surat, GJ"],"demo_name":"Tirupur knitwear dyeing unit"}]}
```

GET `/api/sector/{key}`

Full sector spec: `benchmarks`, `process_steps`, `energy_character`, `regulatory_flags`, `dominant_materials`, and the `demo_profile` used to prefill forms. **404** on unknown key.

GET `/api/reference`

Every emission factor with `value`, `low`, `high`, `unit`, `scope`, `source`, `note` and `typical_price_inr`, grouped by `electricity` / `fuels` / `materials` / `transport` / `waste`. Powers the methodology view; this is the endpoint that makes every number traceable.

GET `/api/corpus`

Flywheel health — how much of each sector benchmark is measured rather than assumed. See [13-DATA-FLYWHEEL.md](#).

```
{"total_plants":9,"total_assessments":9,"prior_strength":8.0,"min_corpus":3,
  "by_sector":[{"sector":"foundry_casting","plants":6,
    "benchmark_weight":0.429,"status":"blended"}],
  "explainer":"Sector benchmarks start as published literature percentiles..."}
```

Anonymous assessment

Stateless. Nothing is written. This is what the public demo hits.

POST /api/assess

```
{ "sector": "textile_dyeing", "state": "Tamil Nadu",
  "annual_output_t": 2400, "annual_revenue_cr": 34, "electricity_kwh": 3000000,
  "fuels": { "COAL_INDIAN": 3900, "DIESEL": 42000 },
  "materials": { "COTTON_CONV": 2650, "PET_VIRGIN": 180 },
  "waste": { "LANDFILL_ORGANIC": 220 }, "freight": { "ROAD_FREIGHT_HCV": 1950000 },
  "eu_export_share_pct": 25, "tariff_inr_per_kwh": 8, "discount_rate": 0.12 }
```

Returns the full assessment (§ *Assessment shape* below). Benchmarked against **literature priors** — the anonymous caller has no plant identity to exclude from the corpus.

GET /api/demo/{sector}

One-call assessment of that sector's built-in demo plant. **This is what the stage demo hits**, so nothing depends on typing.

Authentication

POST /api/auth/register

```
{ "email": "a@b.com", "password": ">8 chars", "name": "Anand Rao",
  "org_name": "Coimbatore Energy Partners", "org_kind": "plant|consultant|corporate" }
```

Creates org + owner user, signs in, sets the cookie. **400** invalid email or password under 8 characters · **409** email already registered.

POST /api/auth/login

`{ "email": "...", "password": "..." }` → sets cookie. **401** on failure, with an **identical message** whether the account is missing or the password is wrong, so the endpoint cannot enumerate registered addresses.

POST /api/auth/logout

Deletes the server-side session row and clears the cookie.

GET /api/auth/me

`{ "authenticated": true, "user": { "user_id": "...", "email": "...", "org_id": "...", "org_name": "...", "org_kind": "..." } }` Never 401s — returns `authenticated: false` for anonymous callers, so the frontend can boot either way.

Plants

All plant routes are **org-scoped at the data layer**. A plant belonging to another organisation returns **404**, not 403 — the existence of the row is itself not disclosed.

GET /api/plants

Each plant with its latest assessment headline and action counts:

```
{
  "plants": [
    {
      "id": "plt_...",
      "name": "Kovai Castings",
      "sector": "foundry_casting",
      "state": "Tamil Nadu",
      "total_tco2e": 18786.4,
      "scope3_tco2e": 12410.2,
      "cp_abatement": 10874.0,
      "cp_benefit_inr": 55080000,
      "actions_done": 1,
      "actions_active": 1,
      "last_assessed_at": 1789...
    }
  ]
}
```

POST /api/plants

`{"name": "Kovai Castings", "sector": "foundry_casting", "state": "Tamil Nadu"}` → the created plant. **400** unknown sector or empty name.

GET /api/plants/{id}

`{"plant": {...}, "assessments": [...], "actions": [...]}` — everything the plant view needs in one call.

DELETE /api/plants/{id}

Soft delete: sets `archived_at`. Assessments and actions are retained; the plant leaves the corpus and the portfolio.

POST /api/plants/{id}/assess

```
{
  "profile": { ...same shape as /api/assess... },
  "label": "FY25 baseline"
}
```

Runs against the **live corpus** (excluding this plant), persists profile + result, and syncs the action rows. Returns the assessment with `assessment_id` and `plant_id` added.

Identity is taken from the plant record, not the profile. `name`, `sector` and `state` are overwritten server-side, so a client cannot rename or re-sector a plant through this endpoint.

Assessments

GET /api/assessments/{id}

The stored result, verbatim as computed. Results are **frozen, never recomputed** — an assessment is a point-in-time statement under a specific set of factors and benchmarks, and an audit trail that silently changes is not an audit trail.

GET /api/assessments/{id}/report

PDF download with the MACC embedded as server-rendered SVG. `?fmt=html` returns the HTML instead. If no browser is available to render, responds with HTML plus an `X-Chakra-Note` header rather than failing.

Actions

PATCH /api/plants/{plant_id}/actions/{intervention_id}

```
{
  "status": "done",
  "act_abatement_tco2e": 3800,
  "act_capex_inr": 1450000,
  "act_annual_benefit_inr": 16100000,
  "notes": "Supplier: Sakthi secondary ingot"
}
```

Status ∈ `recommended` · `planned` · `in_progress` · `done` · `rejected` . **400** otherwise.

`completed_at` is set automatically on entering `done` and cleared on leaving it. Recording actuals is what turns the realisation rate from an assumption into a measurement — every field is optional, and the estimate is used where an actual is absent.

Portfolio

GET `/api/portfolio`

Org-wide rollup — the consultant and corporate view.

```
{
  "plants_total": 9,
  "plants_assessed": 9,
  "total_footprint_tco2e": 188027.3,
  "scope3_share_pct": 66.0,
  "cash_positive_abatement_tco2e": 85455.1,
  "cash_positive_abatement_pct": 45.4,
  "cash_positive_benefit_inr": 521600000,
  "realisation": {
    "actions_total": 178,
    "actions_done": 1,
    "identified_tco2e": 106615.3,
    "committed_tco2e": 3805.0,
    "realised_tco2e": 3800.0,
    "realisation_rate": 0.036,
    "abatement_accuracy": 1.191,
    "capex_accuracy": 0.833,
    "measured_n": 1
  },
  "by_sector": [...],
  "plants": [...]}

```

Assessment shape

Returned by `/api/assess`, `/api/demo/{sector}`, `/api/plants/{id}/assess` and `/api/assessments/{id}`.

headline	total, range, uncertainty %, top leak, peer percentile, cash-positive benefit/capex/payback, plain-English statement
footprint	scope1/2/3, scope_split_pct, biogenic memo line, intensities, grid_source, uncertainty_pct, streams[] { label, scope, tco2e, range, share_pct, activity_qty, activity_unit, source, detail }
sankey	nodes[] {name, kind}, links[] {source, target, value, scope}
leaks	peer_position, leak_count, critical_count, total_gap_to_median_tco2e, benchmark_source, benchmark_provenance, leaks[] { label, rule, severity, share_pct, metric, actual, p50, p75, percentile, finding }
recommendations	count, total_abatement_pct, recommendations[] { name, category, abatement_tco2e, portfolio_abatement_tco2e, derating_pct, lcoa_inr_per_tco2e, capex_inr, net_annual_benefit_inr, payback_months, npv_inr, savings_model, physical_note, caveats, restriction_note, confidence, difficulty, disruption_days }, macc_curve[] { x_start, width, standalone, height, cash_positive }, portfolio { all, cash_positive_only, quick_wins }, blocked[] { name, reason }, assumptions { tariff, discount_rate, derating_note, capex_note }
compliance	cbam { applicable, exposure, basis, caveat }, brsr { readiness[] }
methodology	standard, gwp, factor_note, verification_status, benchmark_note, leak_rule

Field semantics — LCOA, de-rating, savings models, substitution caps — are defined in [11-METHODOLOGY-AND-LIMITATIONS.md](#).

Errors

Code	Meaning
400	Validation — unknown sector, bad status, weak password
401	No session, or expired
404	Not found or not yours
409	Email already registered
422	FastAPI request-body validation

All errors return `{"detail":"..."}` with a message written for the user, not the logs.

Complete Feature Catalogue

Every shipped capability, what it does, and where it lives.  built and verified ·  partial ·  not built.

1. Assessment engine

#	Feature	Status	Where
E1	Scope 1/2/3 inventory per GHG Protocol Corporate Standard		engine/footprint.py
E2	Uncertainty bands (low/base/high) carried through every operation to the headline		engine/factors.py Band
E3	Unit resolution read from each factor's own unit string; unknown units raise rather than silently mis-convert		factors.py _numerator_multiplier
E4	State-level Indian grid factors (15 states, 0.48–0.88 tCO ₂ e/MWh)		factors.py grid_factor
E5	Thermal fuels aggregated to one addressable stream; diesel held separate (it is genset electricity, not process heat)		footprint.py
E6	Biogenic CO ₂ computed as a memo line and excluded from Scope 1		footprint.py
E7	Dual intensity bases — gate-to-gate benchmarked, cradle-to-gate reported only		footprint.py
E8	Stream-level decomposition with share, activity quantity and factor source		footprint.py Stream

2. Leak-point detection

#	Feature	Status	Where
L1	Rule 1 — benchmark breach: intensity above sector p75		engine/leaks.py
L2	Rule 2 — material concentration: >15% of footprint <i>and</i> above median		leaks.py
L3	Rule 3 — structural hotspot: >25% of footprint with no benchmark (catches Scope 3 dominance an energy-only tool misses)		leaks.py
L4	Severity scoring weighted by stream share → critical/high/moderate/watch		leaks.py _severity
L5	Peer percentile by piecewise-linear interpolation, clamped 1–99		leaks.py _percentile
L6	gap_to_median — tonnes recoverable by reaching sector median		leaks.py
L7	One finding per stream even when benchmarked on two bases		leaks.py seen guard
L8	Benchmark override injection + provenance passthrough		leaks.py , assess.py

3. Circular intervention recommender

#	Feature	Status	Where
R1	30-intervention library across energy / material / process / waste / logistics	✓	data/interventions.json
R2	Sector applicability matching	✓	engine/macc.py
R3	Blocking — evaluated then refused with a stated reason (pharma rPET, ceramics briquette)	✓	macc.py BLOCKED
R4	Sector caps — allowed at reduced share with reason (food rPET 35%, auto steel 45%)	✓	macc.py SECTOR_CAPS
R5	Physical blend ceilings per intervention (recycled cotton 25%, alu 70%, steel 80%)	✓	interventions.json max_substitution_pct
R6	Substitution capped at tonnage actually purchased	✓	macc.py _evaluate
R7	Per-card confidence, difficulty 1–5, disruption days, evidence, caveats	✓	interventions.json
R8	Plain-language physical statement of what changes	✓	macc.py _physical_saving_note

4. Economics

#	Feature	Status	Where
C1	Capex annualised by capital recovery factor, never charged wholly to year one	✓	engine/constants.py
C2	$LCOA = (CRF \cdot capex + \Delta opex - saving) / abatement$	✓	macc.py
C3	Five savings models — avoided_purchase, tariff_delta, fuel_switch, price_delta, none	✓	macc.py
C4	Fuel switch correctly returns a net cost where the replacement fuel is dearer	✓	macc.py
C5	Interaction de-rating — each intervention applies to the residual stream in MACC order	✓	macc.py recommend
C6	Standalone <i>and</i> de-rated abatement both reported	✓	macc.py
C7	Simple payback, NPV over asset life, cash-positive flag	✓	macc.py
C8	Banded capex scaling by plant size (not linear)	✓	constants.py size_multiplier
C9	Three portfolios — all, cash_positive_only, quick_wins	✓	macc.py _portfolio_stats
C10	User-overridable tariff and cost of capital	✓	API + UI

5. Compliance

#	Feature	Status	Where
K1	CBAM indicative exposure, apportioned by EU export share	✓	engine/assess.py
K2	CBAM precursor emissions excluded and declared excluded	✓	assess.py
K3	BRSR readiness checklist, explicitly not assurance	✓	assess.py
K4	Per-sector regulatory flags (ZLD, EPR, hazardous waste, fly ash)	✓	data/sectors.json

6. Visualisation

#	Feature	Status	Where
V1	Sankey stream → scope → total, ordered by scope then size to cut ribbon crossing	✓	frontend/app.js
V2	Tail streams bundled into "N smaller sources"	✓	app.js
V3	MACC — width = tonnes, height = ₹/tCO ₂ e, zero line emphasised	✓	app.js
V4	MACC axis clamped to the informative band (8th–92nd percentile × 1.9)	✓	app.js
V5	Off-scale bars dashed + counted, so a clipped bar never reads as its true value	✓	app.js
V6	Leak cards with inline peer-distribution strip (p50, p75, you)	✓	app.js
V7	Expandable working on every recommendation row	✓	app.js detail()
V8	"Considered and rejected" panel	✓	app.js renderRefusals
V9	Zero external dependencies — renders with networking off	✓	no CDN anywhere
V10	Indian number formatting throughout (lakh / crore)	✓	app.js inr()

7. Platform — accounts and persistence

#	Feature	Status	Where
P1	Register / login / logout, bcrypt hashing	✓	backend/auth.py
P2	Server-side revocable sessions (not JWT)	✓	auth.py
P3	Login message identical for wrong password and missing account (no enumeration)	✓	auth.py
P4	Org kinds — plant / consultant / corporate	✓	auth.py , db.py
P5	Org-scoped authorisation enforced at the data layer , not in routes	✓	backend/repo.py
P6	Plant CRUD with soft archive	✓	repo.py
P7	Versioned assessments, results frozen not recomputed	✓	repo.py
P8	Plant identity taken from the record, not the submitted profile	✓	repo.py run_assessment
P9	Denormalised intensity columns for fast corpus queries	✓	db.py
P10	Anonymous sandbox remains fully stateless	✓	app.py

8. Implementation tracking

#	Feature	Status	Where
T1	One action row per recommendation, auto-seeded on assessment	✓	repo.py _sync_actions
T2	Five statuses — recommended / planned / in_progress / done / rejected	✓	repo.py
T3	Estimates refresh on reassessment; status and actuals preserved	✓	_sync_actions
T4	Retires no-longer-applicable interventions <i>unless</i> already acted on	✓	_sync_actions
T5	Actuals capture — abatement, capex, annual benefit, notes	✓	API + completion modal
T6	completed_at set on entering done , cleared on leaving	✓	repo.py update_action
T7	Realisation rate, abatement accuracy, capex accuracy	✓	backend/benchmarks.py
T8	Row order held stable so rows don't jump under the cursor mid-edit	✓	app.js S.actionOrder

9. Data flywheel

#	Feature	Status	Where
F1	Corpus percentiles from real assessments	✓	benchmarks.py
F2	Shrinkage blending toward literature prior at $n/(n+8)$	✓	benchmarks.py
F3	A plant is never benchmarked against itself (enforced in SQL)	✓	_corpus_values
F4	Only the latest assessment per plant counts	✓	_corpus_values
F5	Minimum corpus of 3 before blending begins	✓	benchmarks.py
F6	Provenance returned with every benchmark; UI states which was used	✓	leaks.py + app.js
F7	Corpus health panel — per-sector weight and status	✓	/api/corpus + portfolio

10. Portfolio

#	Feature	Status	Where
O1	Org-wide footprint, Scope 3 share, cash-positive totals	✓	repo.py portfolio
O2	Per-plant rows with latest headline and action counts	✓	repo.py list_plants
O3	Realisation KPI — measured, not assumed	✓	portfolio view
O4	Sector breakdown	✓	repo.py

11. Reporting

#	Feature	Status	Where
Y1	Per-assessment PDF, 3 pages, all 7 sections	✓	backend/report.py
Y2	MACC rendered server-side as vector SVG (no screenshot step)	✓	report.py macc_svg
Y3	Graceful HTML fallback when no browser is available	✓	report.py to_pdf
Y4	Combined 56-page project report from docs/*.md	✓	build_report.py

12. Frontend shell

#	Feature	Status	Where
W1	Hash router — sandbox / auth / portfolio / plant	✓	app.js route()
W2	Auth forms with inline validation errors	✓	app.js initAuth
W3	Plant tabs — Assessment / Action tracker / History	✓	app.js
W4	History with Δ versus previous assessment	✓	app.js renderHistory
W5	Modal system for add-plant, assess and completion capture	✓	app.js modal()
W6	Renderers shared verbatim between sandbox and plant view	✓	one render()

Not built — stated deliberately

Gap	Why it matters	Priority
Full activity-data intake form (fuels, materials, waste, freight)	Reassessment currently edits four fields; the rest come from the sector template	Highest
Email verification, password reset	Account is only as recoverable as the password	High
Rate limiting on auth	Brute force unthrottled	High before public deploy
CSRF token	Mitigated by SameSite=Lax + JSON-only API, not eliminated	High before public deploy
Bill OCR / DISCOM ingest	Manual entry is the real adoption barrier	High
BRSR-format and CBAM working-sheet export	The paywall in the business model	Medium
Vendor marketplace (recommendation → 3 quotes)	Lifts realisation rate	Medium
Multi-user orgs, invites, roles	Consultancies have teams	Medium
Outlier rejection in the corpus	One mis-keyed assessment shifts a small sector	Medium
Schema migrations tool	Second schema change gets awkward	Low now
Session pruning	sessions grows unbounded	Low
ML on realisation outcomes	Needs volume first — see 13-DATA-FLYWHEEL.md §5.2	Later, by design

Counts

API routes	20
Engine modules	6
Platform modules	5
Indian sectors	10
Circular interventions	30
Emission factors	31 + 15 state grids
Detection rules	3
Savings models	5
Refusal mechanisms	3
Tests	20, all passing
Documentation files	19

Team Task Allocation — 4 People

The full build, split four ways, with dependencies, handoffs and a definition of done per task. Written so that four people who have not worked together can start in parallel at hour zero without blocking each other.

The single most important line in this document: the reference data is the product. Code is a few hundred lines anyone competent can write; assembling 30 interventions with defensible abatement fractions, capex bases, savings models and physical ceilings is what a judge cannot replicate by reading the screen. **Role B is the most valuable seat on the team, not the least.**

Roles at a glance

	Role	Owns	Best fit
A	Engine & economics	factors , footprint , leaks , macc , assess	Strongest at maths/logic; comfortable being told their numbers are wrong
B	Domain data & research	emission_factors.json , interventions.json , sectors.json , all citations	Patient researcher; will read a DEFRA table without complaining
C	Frontend	SVG charts, views, router, action tracker	Cares how things look; can hand-roll SVG without reaching for a library
D	Platform, integration & pitch	db , auth , repo , report , API, tests, deck, Q&A	The one who says "no" to scope creep and owns the demo laptop

Pairing rule: A and B must sit together. Every number A computes comes from a file B owns, and every disagreement between them is a bug found early.

Hour-by-hour, by person

H0-H4 · Lock the spine

	Task	Done when
ALL	30 min, whole team: agree "leak point" means benchmark-relative , not sensor-based. Write it on the wall.	Everyone can state it in one sentence
A	Band dataclass with uncertainty arithmetic; FactorDB with unit resolution from the unit string	One quantity × one factor returns a banded tCO ₂ e
B	First 15 factors — electricity and fuels — each with low / base / high , unit , scope , source	JSON parses; every entry has a named source
C	Page skeleton, KPI tiles, CSS variables. No charts yet.	Static page renders with dummy numbers
D	Repo, .gitignore , FastAPI skeleton, /api/health	curl /api/health returns 200

 **Gate H4** — one fuel quantity converts to tCO₂e with a band, through the API.

H4–H12 · Inventory and first sector

	Task	Done when
A	<code>footprint.py</code> — streams, scopes, biogenic memo, both intensity bases	Stream sum equals reported total
B	Finish all 31 factors + 15 state grids. Build the first sector: benchmarks + a realistic demo plant	One sector fully specified
C	Stream table + KPI tiles bound to live API data	Real numbers on screen
D	<code>/api/assess</code> , <code>/api/sector/{k}</code> , <code>/api/demo/{k}</code>	Frontend can fetch an assessment

 **Gate H12** — one real plant profile returns a full Scope 1/2/3 inventory on screen.  **B's trap:** benchmark **only** gate-to-gate (Scope 1+2). Mixing cradle-to-gate and gate-to-gate reported an ordinary plant as catastrophic in our build. Cost four hours.

H12–H20 · Leak detection

	Task	Done when
A	Three detection rules, percentile placement, severity scoring, <code>seen</code> guard	Different rules fire on different sectors
B	Remaining nine sectors with benchmarks and demo plants	10 sectors assess without error
C	Leak cards with the peer-distribution strip (p50 / p75 / you)	A breach is visually obvious
D	Invariant test harness — stream sum = total; no abatement exceeds its stream; all sectors clean	Runs in one command

 **Gate H20** — leaks fire on at least three sectors, with different rules triggering. **D's harness is the most valuable thing built in this block.** Run it after every change from here.

H20–H30 · The MACC — protect this window

Nothing else gets touched until it works. This is the product.

	Task	Done when
A	Target resolution, five savings models, CRF, LCOA, payback, NPV, interaction de-rating	Economics are plausible
B	All 30 interventions — abatement fractions, capex bases, savings models, ceilings, caveats, evidence	Every card has a source and a caveat
C	MACC SVG — clamped axis, clipped-bar marking, rotated labels	Readable from the back of a room
D	Three portfolio views; keep the harness green	<code>cash_positive_only</code> computes

 **Gate H30** — a MACC renders **and the economics are plausible**.  **Plausibility check, do not skip.** Our first run claimed ₹10.5 Cr/yr on a ₹34 Cr plant — 31% of turnover. Three bugs: whole-stream savings applied to tariff-delta and fuel-switch interventions, substitution ignoring its own blend ceiling, and seven electricity interventions stacking to 99% of the meter. **If your cash-positive portfolio exceeds ~15% of revenue, you have one of these.**

H30–H38 · Trust layer

The screens that separate this from a calculator.

	Task	Done when
A	<code>BLOCKED</code> and <code>SECTOR_CAPS</code> . Verify the blocked pairs are actually candidates — ours weren't, so the refusal path silently never fired	<code>blocked[]</code> is non-empty for pharma and ceramics
B	Caveats, confidence ratings, evidence notes on all 30	No card without a caveat
C	"Considered and rejected" panel; expandable working; methodology + limitations section	Refusals visible in one scroll
D	CBAM and BRSR panels	Exposure figure computes with a stated basis

 **Gate H38** — the tool visibly refuses at least two interventions with stated reasons.

H38–H42 · Full stack (if the team is ahead) or polish (if not)

Take this block only if H30 and H38 gates passed on time. Otherwise skip straight to polish.

	Task	Done when
D	<code>db.py</code> schema, <code>auth.py</code> bcrypt + sessions, <code>repo.py</code> org-scoped access	Two orgs cannot see each other's plants
A	<code>benchmarks.py</code> — shrinkage blending, self-exclusion, latest-only	4th plant in a sector flips to <code>blended</code>
C	Router, auth forms, portfolio, plant tabs, action tracker	Full journey clicks through
B	Seed a realistic multi-plant portfolio for the demo	Corpus panel shows blending

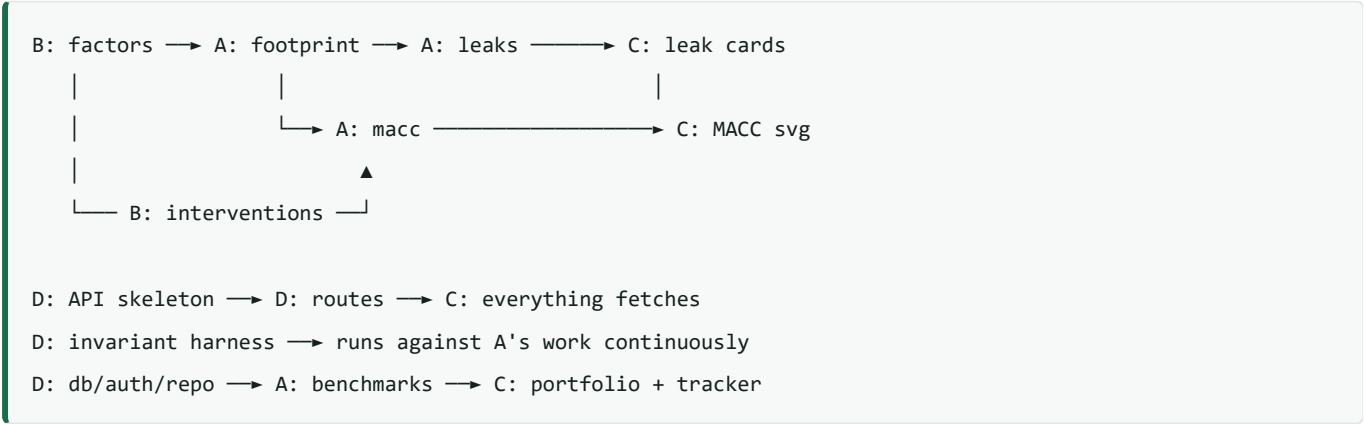
 **Gate H42** — sign up, add a plant, assess, mark an action done, see the realisation rate move.

H42–H45 · Polish and freeze

	Task
C	Sankey ordering (scope then size); Indian number formatting (lakh/crore); <code>hidden</code> beating layout CSS
A	Re-run every sector; confirm zero invariant violations
B	Final pass on citations; make sure every number has a source
D	FEATURE FREEZE. Tag the commit. Build the PDF report.

⚠ An Indian judge reading ₹46,570,000 is doing mental arithmetic instead of listening. Format as ₹4.66 Cr .

	Task
ALL	Full run-through 4x, timed to 3 minutes
ALL	Once with wifi off. If it fails, that is what you just learned
D	Pre-load the hero sector; owns the laptop; writes nothing after freeze
A	Owns methodology Q&A — factors, bands, de-rating, why no ML
B	Owns domain Q&A — why these interventions, where the numbers come from
C	Owns "show me that again" — can navigate anywhere in 2 clicks
D	Owns market Q&A — who pays, why not an existing tool



Nobody is blocked at hour zero. A writes `Band` against a fake factor. B writes JSON needing no code. C builds the shell with dummy data. D builds the API skeleton returning stubs.

A (Engine) - [] Every output carries a band - [] Unknown units raise, never silently convert - [] Stream sum equals total, all 10 sectors - [] No intervention abates more than its stream holds - [] De-rated $\leq$ standalone, always - [] Fuel switch to a dearer fuel shows a **net cost** - [] Can explain LCOA at a whiteboard in 60 seconds

B (Domain) - ☐ Every factor has low / base / high and a named source - ☐ Every intervention has evidence, confidence and at least one caveat - ☐ Every substitution has a physical ceiling with a stated reason - ☐ Benchmarks are gate-to-gate only - ☐ Ten demo plants are realistic, and at least three trigger a breach - ☐ Can defend any single number, or say "that's a literature range, here it is"

C (Frontend) - ☐ Zero external requests — works with networking disabled - ☐ MACC readable from the back of a room - ☐ Sankey ribbons don't cross unnecessarily - ☐ Every recommendation expands to show its working - ☐ Refusals visible without scrolling past the fold - ☐ All money in lakh/crore

D (Platform & pitch) - [] Invariant harness green, one command - [] Two orgs cannot see each other's data (tested, not assumed) - [] `/api/demo/{sector}` returns a full result for the stage demo - [] PDF report generates - [] 3-minute script written and rehearsed 4× - [] Q&A assignments agreed and each person has practised theirs

Cut list — in order, never out of order

When you fall behind, and you will:

Cut	What goes	Safe because
1	Sectors 6–10	Three well-built sectors demo as well as ten
2	Full-stack block (H38–H42)	The engine is the differentiator; accounts are not
3	Sankey	The MACC is the hero; the Sankey is support
4	CBAM / BRSR panels	A good wedge, not the core
5	NPV	Payback and LCOA carry the argument
6	Interventions below 20	Depth beats breadth

Never cut: the MACC · the refusal layer · uncertainty bands · the invariant tests. Those four *are* the differentiation.

If your team is 3, or 2

Three people — merge C and D. One person owns frontend + API + pitch. Drop the full-stack block entirely; ship the stateless engine. This is a perfectly good submission.

Two people — A+B as one (engine and data), C+D as one (everything else). Cut to three sectors and 20 interventions. Keep the MACC and the refusals; they are the whole argument.

Standing rules

1. **Commit after every green invariant run.** Not at the end of a block.
2. **Seed everything.** No live typing during the demo — `/api/demo/{sector}` exists for exactly this.
3. **One person owns the demo laptop** and stops coding on it at freeze.
4. **Write down every assumption as you make it.** At hour 46 nobody will remember why the discount rate is 12%, and a judge will ask.
5. **When A and B disagree about a number, B wins on the source and A wins on the maths.** Resolve it in the JSON, not in a code comment.
6. **Anyone may call a plausibility check at any time.** "That number looks too good" is always a legitimate interrupt.

Security, Tenancy and Testing

Part 1 — Security model

1. Authentication

Decision	Choice	Why
Password hashing	bcrypt with per-password salt	Adaptive cost; salt embedded in the hash; no way to store a password by accident
Session model	Server-side rows , not JWT	Revocable. A user who suspects compromise is logged out everywhere by deleting rows. No key rotation problem. JWTs buy horizontal scale we do not need at a scale we are not at.
Token	<code>secrets.token_urlsafe(32)</code>	256 bits from a CSPRNG
Cookie	<code>HttpOnly</code> , <code>SameSite=Lax</code> , 30-day expiry	JavaScript cannot read it; cross-site POSTs cannot carry it
Minimum password	8 characters	Low bar, honestly stated — see gaps below

1.1 No account enumeration

Login returns an **identical** status and message whether the account does not exist or the password is wrong:

```
if not u or not verify_password(password, u["pw_hash"]):
    raise HTTPException(401, "Email or password is incorrect.")
```

Registration necessarily reveals that an email is taken (409). That is unavoidable for a self-serve signup and is the standard trade.

Tested: `test_wrong_password_rejected_without_revealing_account` asserts both responses are byte-identical.

2. Tenancy — the important one

Authorisation is enforced at the data layer, not in route handlers.

Every function in `repo.py` that touches plant data takes an `org_id` and scopes its query by it:

```
def get_plant(org_id: str, plant_id: str) -> dict:
    p = db.one("SELECT * FROM plants WHERE id=? AND org_id=? AND archived_at IS NULL",
               (plant_id, org_id))
    if not p:
        raise HTTPException(404, "Plant not found.")
```

This matters because the common failure mode is a route that forgets its check. Here, **there is no query in the module that would return another organisation's rows** — a forgetful route cannot leak data, because the function it calls cannot produce it.

2.1 404, not 403

A plant belonging to another org returns **404 Not Found**, never 403 Forbidden. 403 confirms the row exists, which is itself a disclosure. An attacker enumerating plant IDs learns nothing.

2.2 Identity cannot be hijacked through the payload

`POST /api/plants/{id}/assess` accepts a profile, but overwrites its identity fields from the plant record:

```
profile["sector"] = plant["sector"]
profile["name"]   = plant["name"]
```

Without this, a client could re-sector a plant through the assessment endpoint — which would also poison the benchmark corpus for a sector it does not belong to. **Tested:**

`test_plant_identity_cannot_be_hijacked_via_profile` .

3. Privacy posture

Property	Status
Anonymous sandbox stores nothing	✓ Tested — assessment count unchanged after <code>/api/assess</code>
Plant data never leaves the request in sandbox mode	✓ No persistence path
No third-party calls at runtime	✓ Zero external requests, by design
No analytics, no trackers, no CDN	✓ Frontend has no external dependencies
Corpus is aggregate-only	✓ Percentiles across a sector; no plant identity reachable from a benchmark
A plant's own data excluded from judging it	✓ Enforced in SQL

The one-sentence answer to "where is my data stored?" — in the sandbox, nowhere; signed in, in your own organisation's rows, and the only thing that ever leaves them is an anonymous contribution to a sector percentile. This matters commercially, not just ethically. Energy and material consumption is competitively sensitive; an SME that believes its data is pooled with a rival's will not enter it.

4. Known gaps — stated, not hidden

#	Gap	Risk	Mitigation today	Fix
S1	No rate limiting on auth	Brute force unthrottled	bcrypt cost makes it slow, not impossible	Per-IP limiter before public deploy
S2	No CSRF token	Cross-site state change	<code>SameSite=Lax</code> + JSON-only API + no form posts	Double-submit token
S3	No email verification	Signup with someone else's address	None	Verification link
S4	No password reset	Account unrecoverable	None	Emailed token
S5	Sessions never pruned	<code>sessions</code> table grows	Expiry checked on read	Periodic sweep
S6	Single-org users	No teams, no roles beyond <code>owner</code>	None	Invites + roles
S7	SQLite single writer	Contention under load	Fine at current scale	Postgres
S8	No audit log	Cannot see who changed an action	<code>updated_at</code> only	Append-only log
S9	Errors may leak detail	Minor information disclosure	Messages are user-facing, not stack traces	Error taxonomy

None of these block the demo. S1 and S2 block a public deployment. Saying so is more useful than a security section that implies completeness.

Part 2 — Testing

5. What is tested

`prototype/backend/tests/test_stack.py` — **20 tests, all passing**, against a throwaway SQLite file so the working database is never touched.

```
cd prototype/backend
python -m pytest tests/test_stack.py -q
```

Area	Tests	What they prove
Anonymous surface	3	Sandbox persists nothing; demo needs no auth; protected routes 401
Authentication	5	Register/login/logout cycle; duplicate rejected; wrong password indistinguishable from missing account; weak password rejected; password never stored in plaintext
Tenancy	1	Org B cannot list, read, assess or delete Org A's plant
Plants & assessments	3	Assessment persists and round-trips; identity cannot be hijacked via profile; unknown sector rejected
Action tracking	2	Actions seeded from recommendations; status and actuals survive reassessment ; invalid status rejected
Data flywheel	4	Blending begins only at sufficient n and with the correct weight; a plant is never benchmarked against itself; only the latest assessment per plant counts ; corpus endpoint reports provenance
Portfolio	1	Rollup sums correctly; realisation moves when an action completes
Reporting	1	HTML and PDF render; MACC embedded as vector, not omitted

5.1 The tests that matter most

Three assertions carry disproportionate weight, because each guards a property that would be invisible if it broke:

```
# A plant that benchmarks against itself drags its own percentile toward its
# own value. The detector goes quiet and the product silently stops working.
assert n_without == n_with - 1

# A reassessment that resets someone's tracker destroys months of their work
# without an error message.
assert after[iid]["status"] == "done"
assert after[iid]["act_abatement_tco2e"] == 42.0

# One plant assessed twelve times must not outvote eleven plants assessed once.
assert len(vals) == 1
```

6. Engine invariants

Checked across all 10 sectors on every run, independent of the unit tests:

Invariant	Why it would be invisible
Stream emissions sum to the reported total	A dropped stream just makes the number smaller; nothing errors
No intervention abates more than its target stream contains	Produces a plausible-looking oversized recommendation
De-rated abatement never exceeds standalone	Silently inflates the portfolio
Substitution never exceeds its declared blend ceiling	Recommends something the process cannot physically accept

Current result: 0 violations across 10 sectors.

7. Manual verification performed

Beyond automated tests, the full journey was driven in a real browser:

- Registered a consultancy account through the actual form
- Created 9 plants across 3 sectors and assessed each
- **Watched the flywheel flip** — plants 1–3 benchmarked against literature, plant 4 onward `blended`, confirming self-exclusion is wired in
- Marked an action `planned`, another `done` with actuals recorded
- Confirmed realisation moved from `null` → 3.6%, abatement accuracy 119%, capex accuracy 83%
- Generated the per-assessment PDF: 3 pages, all 7 sections, MACC present on page 2

7.1 Bugs found by manual testing that the unit tests missed

Bug	Why tests missed it
<code>.modal{display:flex}</code> beat the <code>hidden</code> attribute, so the modal showed on page load	CSS specificity — no API test can see it
Sector names rendered as raw keys on a direct load to <code>/portfolio</code>	State only populated by a different view
Action rows re-sorted the instant a dropdown changed, jumping under the cursor	Correct data, bad interaction

All three are UI-layer, which is exactly the class of defect an API test suite cannot reach. **A green test suite is not a working product** — it is a working backend.

8. Not tested

Gap	Consequence
No frontend unit tests	Chart maths is verified by eye and by DOM inspection, not by assertion
No load or concurrency testing	SQLite write contention is untested
No property-based testing on the engine	Edge cases in band arithmetic could hide
No browser-automation regression suite	UI bugs can reappear silently
No security scanning or dependency audit	Vulnerable transitive dependency would go unnoticed

At 48 hours these are correct omissions. At a real deployment, the frontend regression suite is the first to add — it is where all three manually-found bugs lived.